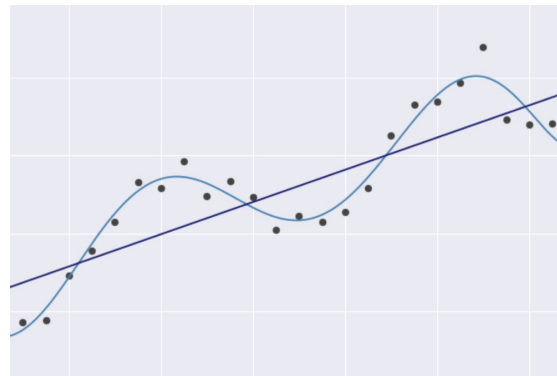




CLASS 1 - INTRODUCTION

What is Analytics?

Using **data** to build **models** that lead to better **decisions** that **add value** to an organization's mission



Note: Analytics can be both “big data” and “small data”

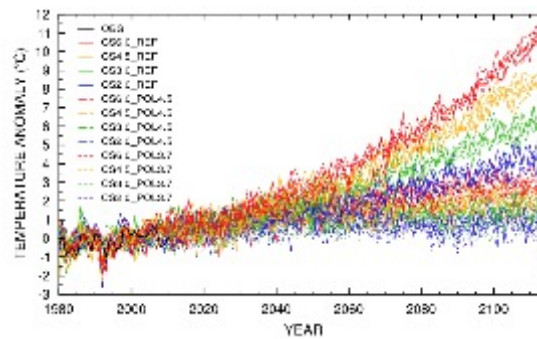
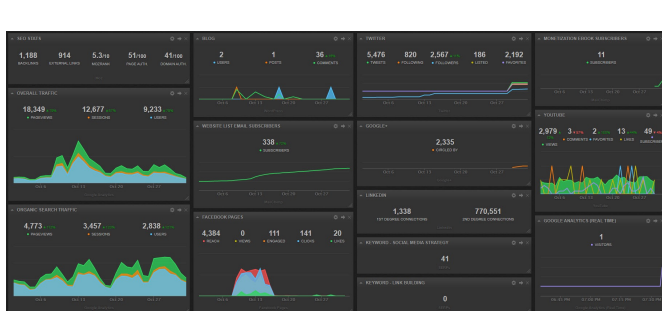
From Data to Decisions

Descriptive Analytics: patterns in the data

Predictive Analytics: insight into outcomes

Prescriptive Analytics: business decisions

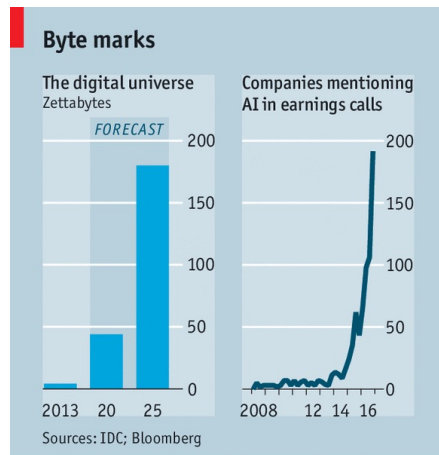
- Data management
- Machine learning
- Optimization
- Data visualization
- Forecasting



Why Now?

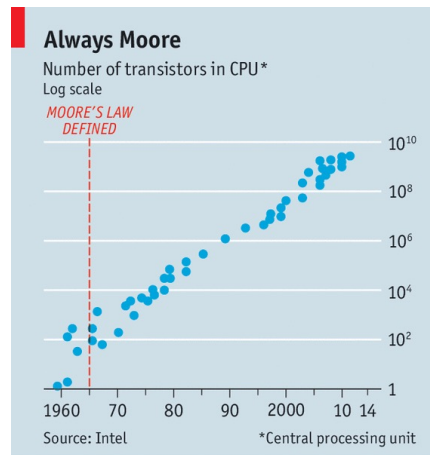
- Analytics is not new — it has been around for decades
- But today's environment creates new opportunities to transform business practices through analytics

Increased availability and lower cost of data



Economist.com

Unprecedented computing power



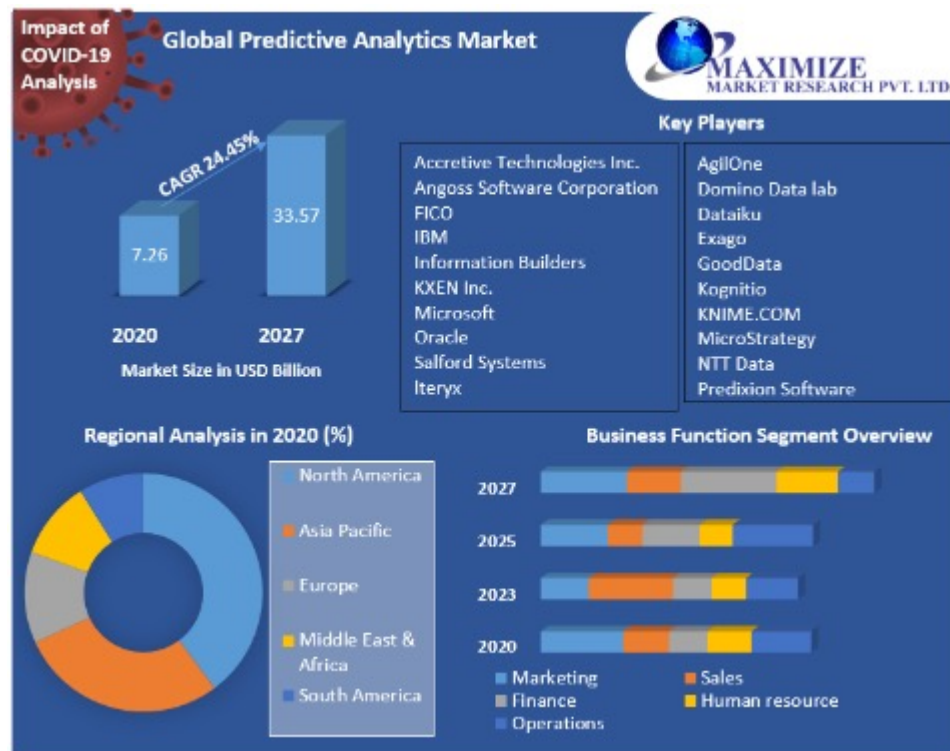
Economist.com

Analytics-savvy business leaders



Growth of Analytics

- Extremely strong growth in demand for analytics professionals
- And analytics is critical component of top MBA program value proposition



Source: Maximize Market Research, PVT. LTD.

Age of AI

- Do we still need to learn analytics in the age of AI?
- My reasons:
 - Analytics easier with AI help
 - Understand fundamentals behind AI's “end-to-end” solution
 - Small data environment
 - Basics to better understand the world and solve problems
 - “Matrix” and human agency

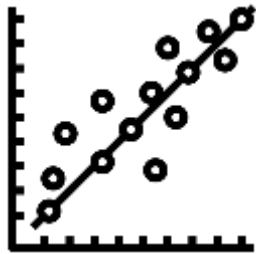
Motivation, Philosophy, Objectives

Course Objectives

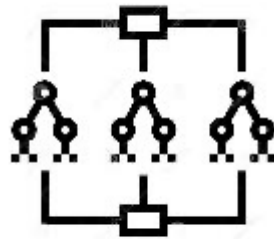
- Identify competitive advantages to individuals and companies
- Learn the benefits (and limitations) of analytics techniques
 - power and importance of data
 - how analytics methods work
 - which methods are most salient in which context
 - how to understand, interpret, use, and implement analytics results
- Communicate analytics to other stakeholders
- Inspire you to leverage analytics in your career and within your organization
- Overarching goal:

improve/ change the way you think, with data and models

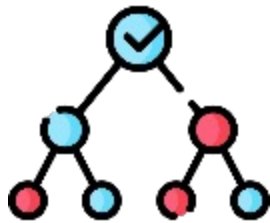
Predictive Analytics



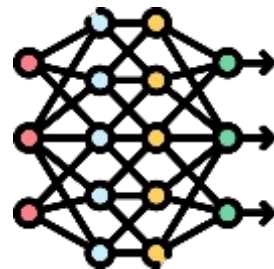
Linear Models



Ensemble Methods



Tree-Based Models

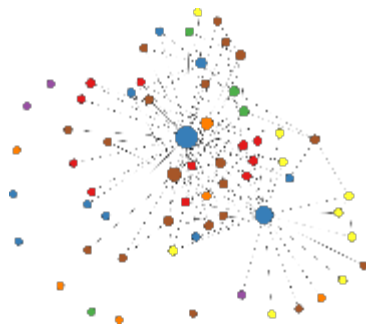


Deep Learning

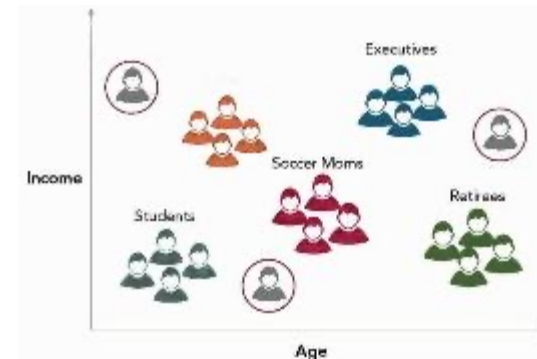


Collaborative Filtering

Descriptive Analytics



Network Analysis

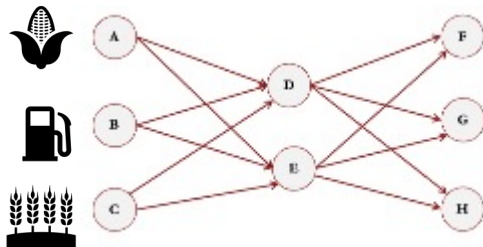


Clustering

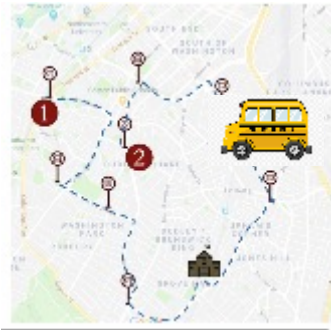


Text Analytics

Prescriptive Analytics



Linear Optimization
World Food Programme



Multi-Objective Optimization
School Bus Routing



Integer Optimization
Sponsored Ads and
Supermarket Promotion



Stochastic Optimization
Power Grid

Sample Application Areas

- solar power generation
- customer retention
- heart disease
- parole violators
- medical costs
- sponsored ad click-through rates
- housing prices
- hospital readmissions
- customer segmentation and CRM
- contract design
- stock returns
- Trump tweets
- online reviews
- social networks
- retail recommender systems
- optimization in supply/demand chains
- online advertising
- public school busing
- landfill siting and optimization
- ...

→ An ever-growing list of applications across functions and industry sectors



With opportunities come potential challenges...

With opportunities come potential challenges...

- Challenges: bad data, bad models, inappropriate uses, spurious relationships, bias and fairness issues....
- Learning new concepts and new modeling tools + how to navigate, avoid, identify, and mitigate the above challenges
- This is also part of “changing the way you think” with data and models

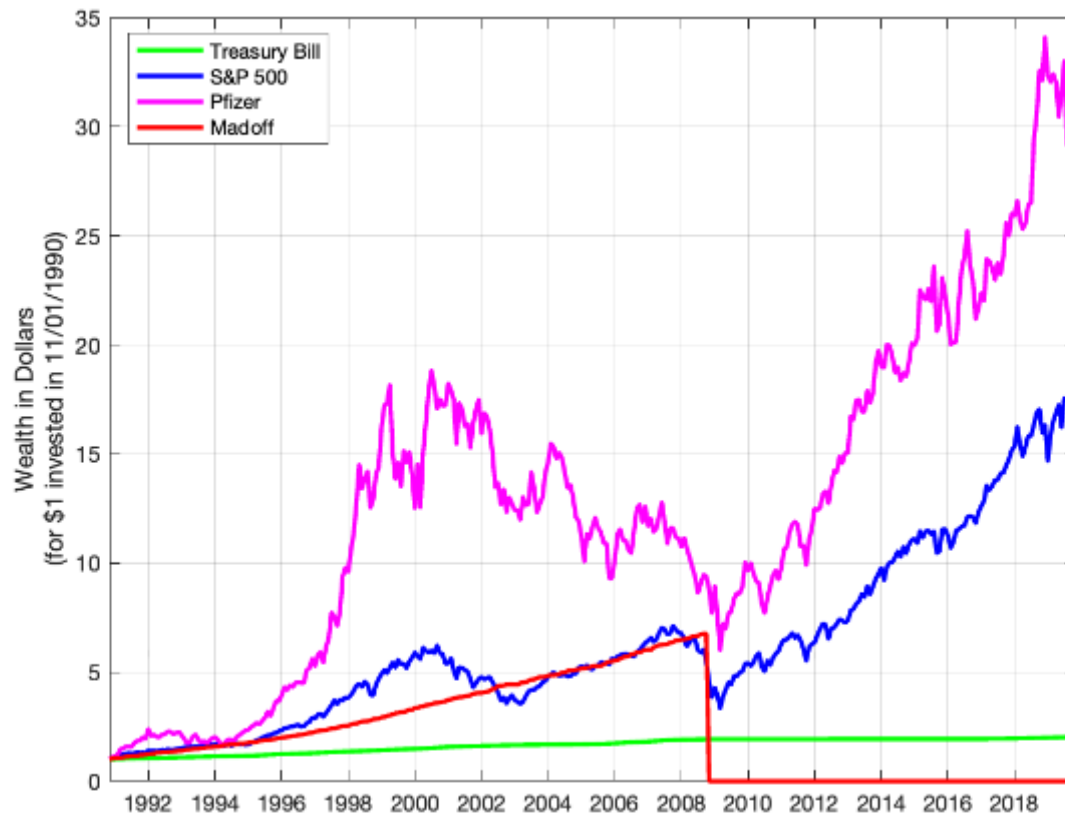
Correlation can be different from causation

Year	Electoral vote result ^[10]	Game score	Popular vote winner	Incumbent popular vote	Rule upheld?
1932	Roosevelt defeats Hoover 472–59	19–6 win	Roosevelt	Won electoral vote	Yes
1936	Roosevelt defeats Landon 523–8	13–10 win	Roosevelt	Won electoral vote	Yes
1940	Roosevelt defeats Willkie 449–82	37–10 win	Roosevelt	Won electoral vote	Yes
1944	Roosevelt defeats Dewey 432–99	14–10 win	Roosevelt	Won electoral vote	Yes
1948	Truman defeats Dewey & Thurmond 303–189–39	59–21 win	Truman	Won electoral vote	Yes
1952	Eisenhower defeats Stevenson 442–89	23–24 loss	Eisenhower	Lost electoral vote	Yes
1956	Eisenhower defeats Stevenson 457–73	20–9 win	Eisenhower	Won electoral vote	Yes
1960	Kennedy defeats Nixon 303–219	10–31 loss	Kennedy	Lost electoral vote	Yes
1964	Johnson defeats Goldwater 486–52	27–20 win	Johnson	Won electoral vote	Yes
1968	Nixon defeats Humphrey & Wallace 301–191–46	10–13 loss	Nixon	Lost electoral vote	Yes
1972	Nixon defeats McGovern 520–17	24–20 win	Nixon	Won electoral vote	Yes
1976	Carter defeats Ford 297–240	7–20 loss	Carter	Lost electoral vote	Yes
1980	Reagan defeats Carter 489–49	14–39 loss	Reagan	Lost electoral vote	Yes
1984	Reagan defeats Mondale 525–13	27–14 win	Reagan	Won electoral vote	Yes
1988	H. W. Bush defeats Dukakis 426–111	27–24 win	G. H. W. Bush	Won electoral vote	Yes
1992	B. Clinton defeats G. H. W. Bush 370–168	7–24 loss	B. Clinton	Lost electoral vote	Yes
1996	B. Clinton defeats Dole 379–159	31–16 win	B. Clinton	Won electoral vote	Yes
2000	G. W. Bush defeats Gore 271–266	21–27 loss	Gore	Lost electoral vote	Yes
2004	G. W. Bush defeats Kerry 286–251	14–28 loss	G. W. Bush	Won electoral vote	No
2008	Obama defeats McCain 365–173	6–23 loss	Obama	Lost electoral vote	Yes

From 1936-2008, the incumbent party results of the US presidential elections were nearly perfectly correlated with the results of the last home game of the Washington D.C. (American) football team.

https://en.wikipedia.org/wiki/Redskins_Rule

The past might not be a good predictor of the future...

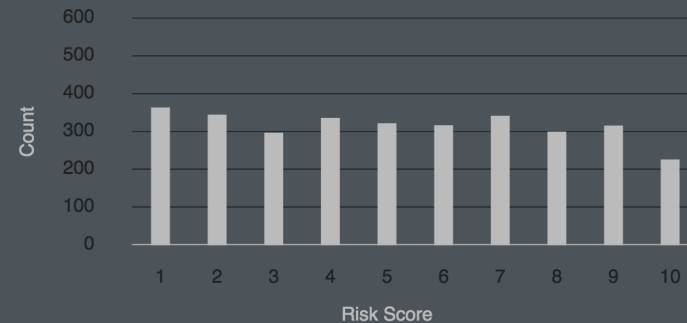


Courtesy of Prof. Adrien Verdelhan of MIT Sloan

Analytics models can exhibit bias/unfairness



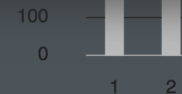
Black Defendants' Risk Scores



White Defendants' Risk Scores

Rating systems may discriminate against Uber drivers

By Leslie Morris | December 15, 2016



One might be studying the wrong problem (or the right problem with the wrong data)

- In WWII, German antiaircraft were responsible for downing a large number of British RAF planes.
- British military Operations Research unit mapped bullet holes to determine where to put extra armor on aircraft



One might be studying the wrong problem (or the right problem with the wrong data)

- In WWII, German antiaircraft were responsible for downing a large number of British RAF planes.
- British military Operations Research unit mapped bullet holes to determine where to put extra armor on aircraft

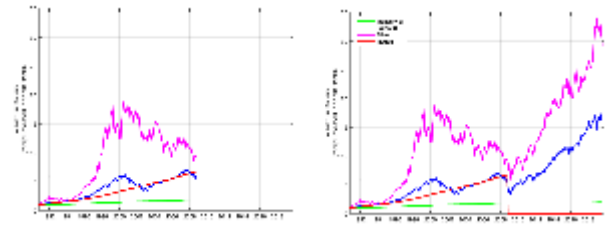


**But the data was from planes that
were not shot down**

With opportunities come potential challenges...

Flawed data methods

- Data look backward, decisions look forward
- Solutions: Field experimentation, “what-if” simulation, human judgment



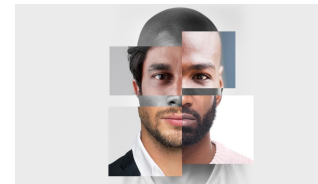
Spurious relationships

- Coincidental correlations
- Correlations due to confounding factors
- Solutions: Econometrics, human judgment

Year	Electoral vote result	Washington Football Team Win or Lose?	Incumbent Party Win or Lose?
1936	Roosevelt defeats Landon 523–8	win	win
1940	Roosevelt defeats Willkie 449–82	win	win
1944	Roosevelt defeats Dewey 432–99	win	win
1948	Truman defeats Dewey 303–189–39	win	win

Bias/Fairness Issues

- Analytics-based decisions with disparate impacts on sub-populations
- Solutions: “Fair” analytics, human judgment



Wrong problem or model

- Data scarcity, missing data, inaccurate data
- Inadequate model fitting, model overfitting
- Solutions: Model validation, human judgment



With opportunities come potential challenges...

- Challenges: bad data, bad models, inappropriate uses, spurious relationships, bias and fairness issues....
- In addition to teaching you new concepts and new modeling tools to add value to you and your enterprises, we also will aim to show you how to navigate, avoid, identify, and mitigate the above challenges
- This is also part of “changing the way you think” with data and models



Okay, let's start learning some analytics...

Google, Search-engine Arithmetic, and Network Analytics

Web Search in the 1990's

- Let's go back to the 1990s, the beginnings of the internet.
- Web search was a new problem with many competitors.
 - Ask Jeeves, Altavista, Lycos, Yahoo!

- Search results quality was questionable.
- Google came along in 1998 with an innovative algorithm called PageRank.



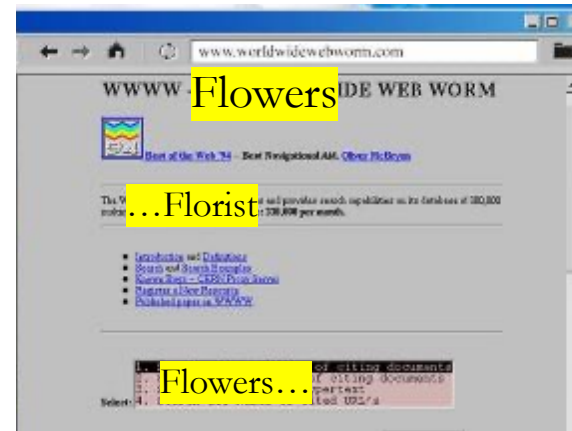
Let's Design a Method to Rank Webpages

- What are some straightforward methods to rank webpages?
- How would you go about ranking those pages in terms of relevance?
 - Thought experiment: search for “flowers”
- Assume you have the entire content information about each page: text, images, videos, hyperlinks, etc.
- Try to prescribe a way to determine a numerical score that is higher if the page is more relevant to the query (in this case, “flowers”)

Simple Ranking Algorithm



Query
“Flowers”



Website

- Assign numerical weights w_1 , w_2 , w_3 to a query-website combination
 - w_1 is the weight if the query is in the title of the page
 - w_2 is the weight of number of times the query appears on the page
 - w_3 is the weight of number of times similar words (e.g., florist) appear
- Add up the weighted value of each page and rank accordingly. The method is simple and intuitive! But what's the problem?

Google's Innovative Idea

- **Key Idea:** Filter websites by keywords, but rank based on how “popular” or “influential” a website is.

The New York Times



vs.

Some random
internet blog

- How to measure “popularity?”
- Useful information from hyperlinks...

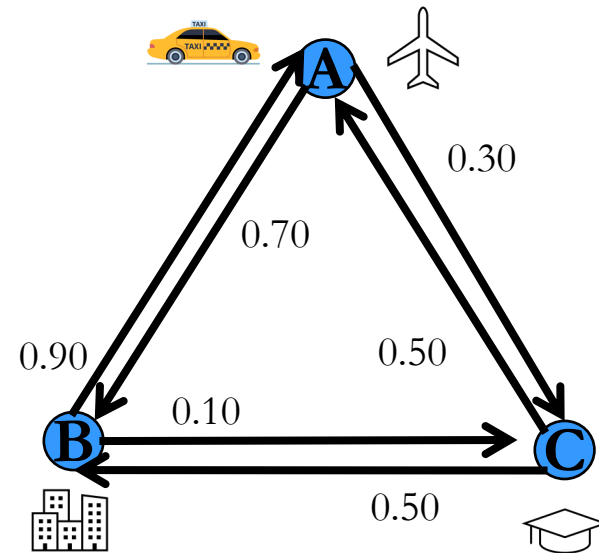
Let's take a quick detour...

- Let's go to the world of taxi/Uber/Lyft drivers



Random Driver Model

		Next Location		
		A 	B 	C 
Current Location	A 		0.70	0.30
	B 	0.90		0.10
	C 	0.50	0.50	



- Given the “**Current Location**”, the next passenger brings the taxi to “**Next Location**” with certain probabilities.
- Question:** After 50 trips, where is the taxi likely to be? (A, B or C)

Computing Likelihoods

Likelihood of being at Current Location	Current Location	Next Location:		
		A	B	C
.25	A		.70	.30
.25	B	.90		.10
.50	C	.50	.50	
Likelihood of Next Location:		$.25 \times .90 + .50 \times .50$ = .475	$.25 \times .70 + .50 \times .50$ = .425	$.25 \times .30 + .25 \times .10$ = .10

Computing Likelihoods

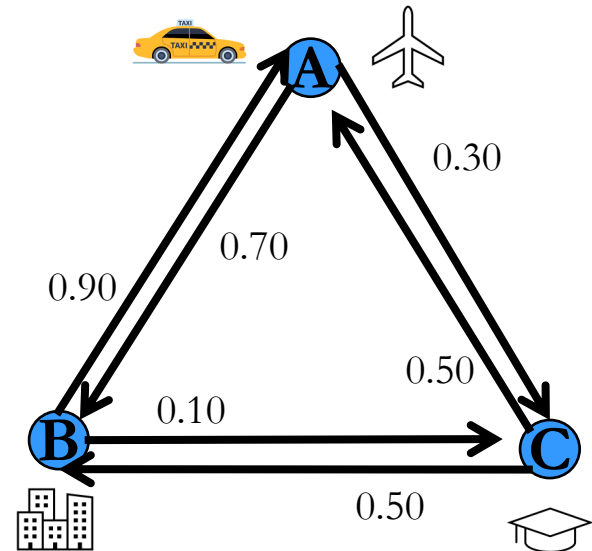
Likelihood of being at Current Location	Current Location	Next Location:		
		A	B	C
P_a	A		.70	.30
P_b	B	.90		.10
P_c	C	.50	.50	
Likelihood of Next Location:		$P_b * .90 + P_c * .50$	$P_a * .70 + P_c * .50$	$P_a * .30 + P_b * .10$

Probability Distribution of Visited Locations

Start at location **A**, compute the probability of being at locations A, B, and C after 1, 2, 3, ... trips:

Location:

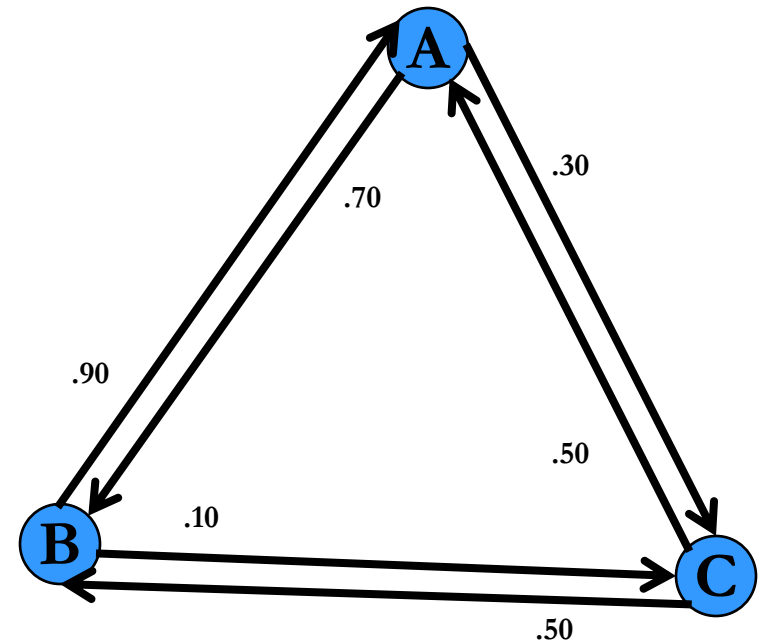
		A	B	C
Number of Trips	0	1	0	0
	1	0	0.700	0.300
	2	0.780	0.150	0.070
	3	0.170	0.581	0.249
	4	0.647	0.244	0.109
	...	...	...	...



Probability Distribution of Visited Locations

Start at location **A**, compute the probability distribution of being at locations A, B, and C after 1, 2, 3, ... trips:

		Location:		
		A	B	C
Number of Trips	1	0	0.700	0.300
	2	0.780	0.150	0.070
	3	0.170	0.581	0.249
	4	0.647	0.244	0.109
	5	0.274	0.508	0.219
	...	...	...	...
	50	.438	.392	.170
	51	.438	.392	.170



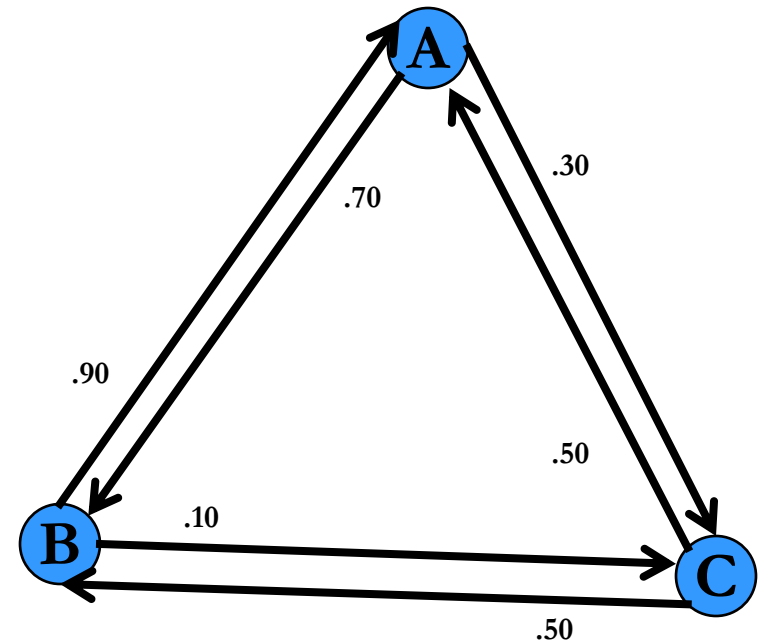
Probability Distribution of Visited Locations

Start at location **C**, compute the probability distribution of being at locations A, B, and C after 1, 2, 3, ... trips:

Location:

Number of Trips

	A	B	C
1	0.500	0.500	0
2	0.450	0.350	0.200
3	0.415	0.415	0.170
4	0.459	0.376	0.166
5	0.421	0.404	0.176
...	...	...	...
55	.438	.392	.170
56	.438	.392	.170



Probability Distribution of Visited Locations

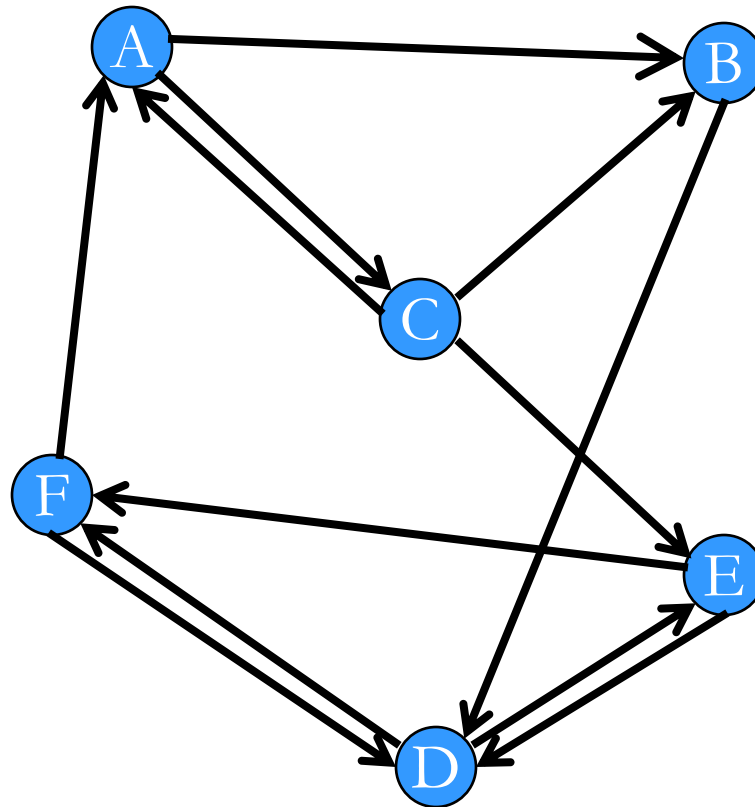
- Regardless of where a driver starts, eventually (after around 55 trips) the probability distribution of her/his current location will be:

	Location		
	A	B	C
Probability	.438	.392	.170

- This is a property of a probabilistic system known as a Markov chain (which is typically taught in first graduate course in probability theory)
- This also means that with a fleet of 1000 cars, eventually:
 - ~438 drivers will be at location A
 - ~392 drivers will be at location B
 - ~170 drivers will be at location C
 - Location A appears more “popular” than B and C

World Wide Web as a Network

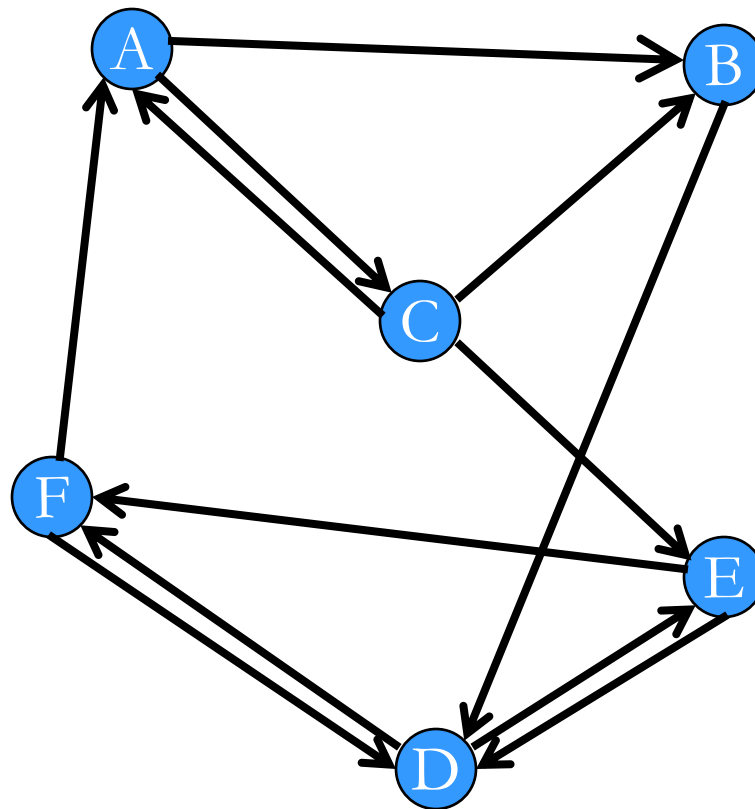
- Vertices are pages
- Arcs are hyperlinks



Web User Model

Model: users go from page to page with equal probability among all hyperlinks from their current page

If I start at page C, I next go to page A, B, or E each with probability $p = 0.333$



Web Network Model, continued

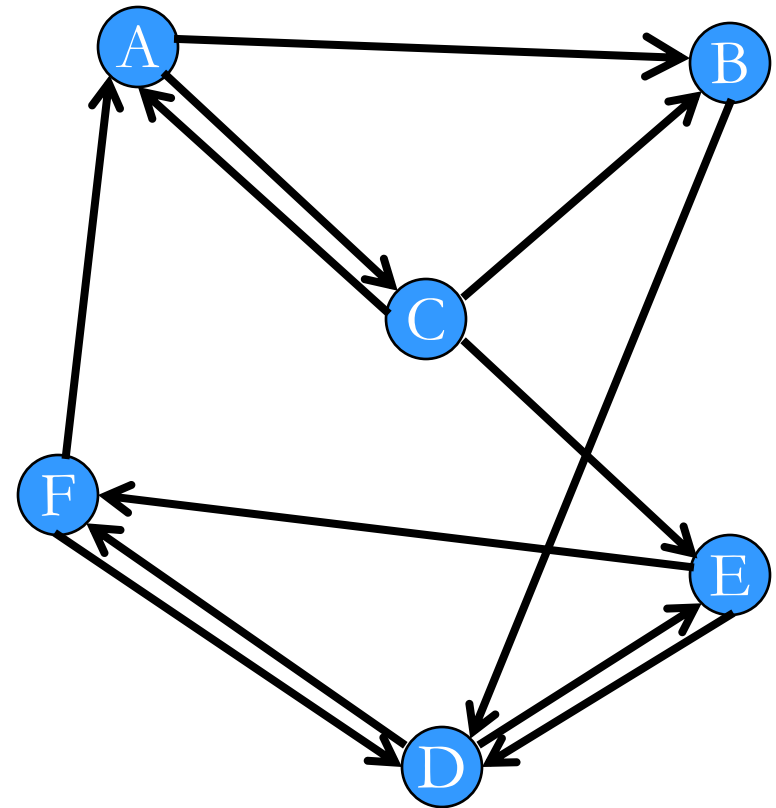
- Think of a hyperlink as a recommendation: a hyperlink from my homepage to yours is my endorsement of your webpage
- A page is more important/valuable/influential if it receives many recommendations
- But the stature of the recommender also matters...

Model Probability Table

Next Page:

Current Page

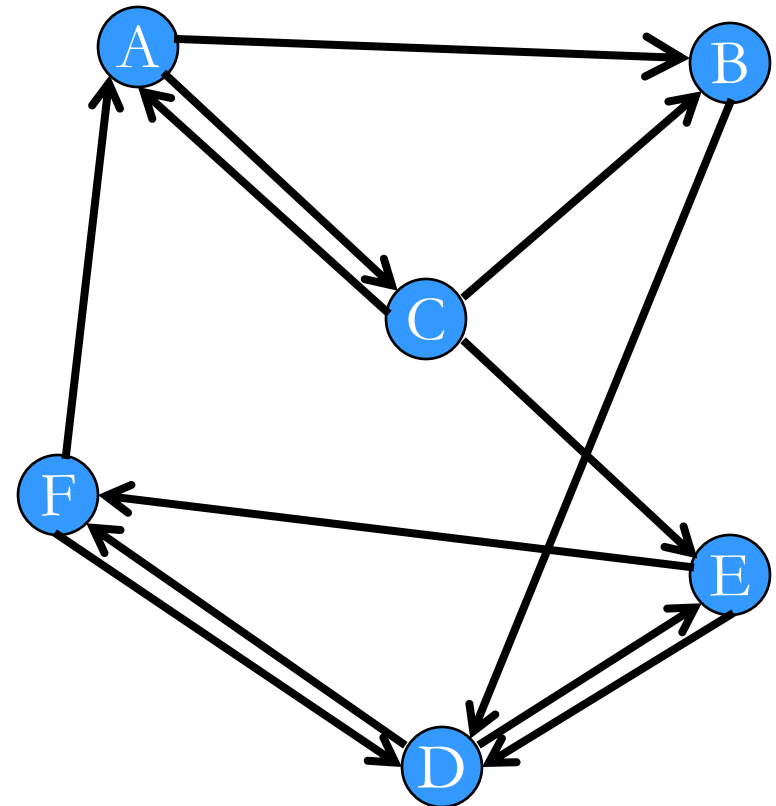
	A	B	C	D	E	F
A		.5	.5			
B				1.0		
C	.333	.333			.333	
D					.5	.5
E				.5		.5
F	.5			.5		



Probability Distribution of Visited Pages

Start on page **A**, compute the probability distribution of visited pages after 1, 2, 3, ... click-thrus:

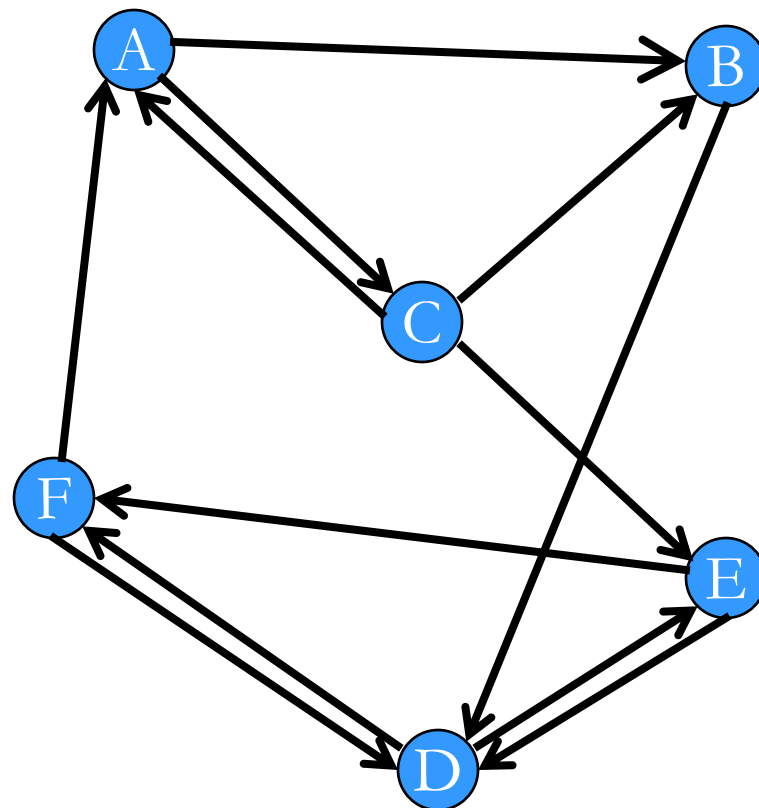
		Page:					
		A	B	C	D	E	F
Number of Click-Thrus	1	0	0.5	0.5	0	0	0
	2	0.167	0.167	0	0.5	0.167	0
	3	0	0.083	0.083	0.25	0.25	0.333
	4	0.194	0.028	0	0.375	0.153	0.25
	5	0.125	0.097	0.097	0.229	0.188	0.264
	...	...	...	...	...	...	...
	67	.140	.093	.070	.295	.171	.233
	68	.140	.093	.070	.295	.171	.233



Probability Distribution of Visited Pages

Start on page **E**, compute the probability distribution of visited pages after 1, 2, 3, ... click-thrus:

Page:		A	B	C	D	E	F
Number of Click-Thrus	1	0	0	0	0.5	0	0.5
	2	0.25	0	0	0.25	0.25	0.25
	3	0.125	0.125	0.125	0.25	0.125	0.25
	4	0.167	0.104	0.063	0.313	0.167	0.187
	5	0.115	0.104	0.083	0.281	0.177	0.240
	...	...	...	...	...	...	...
	63	.140	.093	.070	.295	.171	.233
	64	.140	.093	.070	.295	.171	.233



Probability of Visited Pages as Popularity Measure

- Regardless of where a user starts, eventually (after around 70 visits) the user's probability distribution of visited pages will be:

	Page					
	A	B	C	D	E	F
Probability	.140	.093	.070	.295	.171	.233

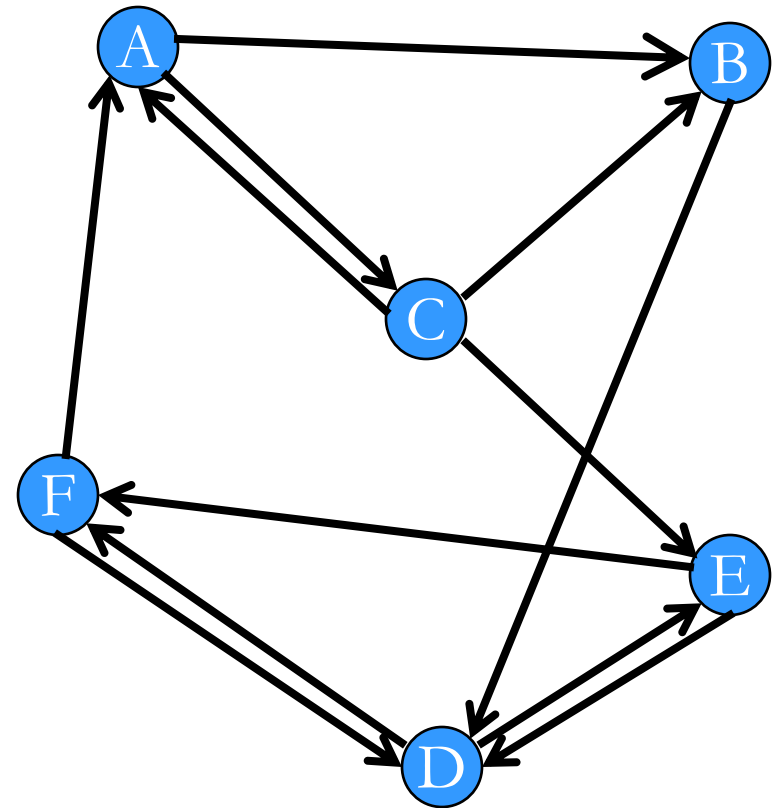
- What about all users? 14% of all users will be visiting page A, 9.3% will be visiting page B, ..., 23.3% will be visiting page F
- Again, this is a property of a system known as a Markov chain (that is typically taught in a first graduate course in probability theory)

Model Probability Table

Next Page:

Current Page

	A	B	C	D	E	F
A		.5	.5			
B				1.0		
C	.333	.333			.333	
D					.5	.5
E				.5		.5
F	.5			.5		



Probability Distribution and Page Ranking

- The probability that a page is visited is equivalent to its importance/relevance/influence and hence is used for ranking:

	Page					
	A	B	C	D	E	F
Probability	.140	.093	.070	.295	.171	.233
“PageRank”:	.140	.093	.070	.295	.171	.233

- This is the essence of the PageRank methodology developed by Google

Web Network Model, cont.

- Think of a hyperlink as a recommendation, or an endorsement
- A page is more important/valuable/influential if it receives **more recommendations**
- A page is more important/valuable/influential if it receives **recommendations from other influential pages**
 - An endorsement from Jeff Bezos probably does more to boost a job application than 20 endorsements from 20 unknown colleagues
 - However, if Jeff Bezos is very generous with praise, then his endorsements drop in importance/value/influence
- PageRank implicitly accounts for these considerations across the entire network

PageRank Innovation

- Brin and Page proposed the idea of ranking pages using a minor variant of this method
 - Page filed for a patent in 1998 (the patent was granted in 2001)
- Method is “easy” to compute
 - Just do the simple arithmetic (for 30 billion pages)
 - Challenging from a computational engineering point of view
 - But a simple concept based on very elementary probability

PageRank Innovation, continued

- Many enhancements to PageRank, including:
 - Users typing in a new URL, spending more or less time on each page, clicking on hyperlinks with higher or lower likelihood, etc.
 - Personalization of rankings (for you) through data analytics and machine learning (PageBrain)
 - Modifications for e-commerce applications
- PageRank idea is used extensively today in public relations, college rankings, board/committee memberships, social networks, etc.
- PageRank plays a role when we discuss **Social Network Analytics** later today...

Google's PageRank

- Simple but powerful use of basic probability concepts
- Many other analytics developments at Google
 - Prediction of Click-Thru Rates (CTRs)
 - Text analytics, auto-complete, and predictions
 - Online advertising optimization (Google AdWords)
 - Social Network Analytics
- Google's integration of analytics techniques has resulted in one of the world's most successful companies

Course Administration

Course Logistics

- Class Sessions Schedule

	Day(s)	Time	Location
Lecture	Monday/Thursday	9:00-11:15am	Rand
TA Session	Friday	?	

- Grading

Components	Percent
Individual Assignments	50%
Team Project	35%
Class Participation	15%

Attendance

- Lecture attendance is required
 - Missing one class session does not carry penalty
 - Each additional missed class reduces total grade by 5%
 - Automatic failure if missing 4 or more class sessions
- TA sessions: Q/A for homework assignments
- I will also hold office hour from 2pm to 3pm on every Wednesday

Class Participation

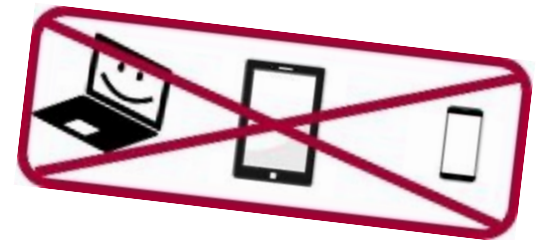
- Class participation is very important
 - Allows fruitful exchange of ideas/observations
 - Ask questions about material you do and do not understand
 - Demonstrates your understanding of class material
 - Helps everyone engage
 - Helps us learn from each other!
- Class participation helps me gauge how the class is keeping up (or not)
- Class attendance: you can't participate if you're not in class...
- **Please use your name tag in class**

Honor Code and Classroom Norms

We take integrity and its violations very seriously

Specific Rules of this Classroom

- No laptops open or Cellphone in class
- You may take notes on a tablet device
- **Your class participation score will decrease steeply if you are observed using your phone or if your laptop is open**



Time Matters

- Late arrivals are disruptive to everyone and carry penalties in grading
- Please use Fuqua Checkin to sign in your seat every time you arrive to class.

Individual Assignments

- Weekly assignments generally due on Sunday at 11:59pm on Canvas.
- Assignments consist of a Notebook in R or Python.
 - 1-2 questions, each with multiple parts that guide you through an analysis.
 - You will need to write some code.
 - Provide written answers directly in the notebook.
- Collaboration Policy
 - **Allowed:** discussing course concepts with fellow students, seeking debugging assistance (including with AI)
 - **Disallowed:** providing or copying solutions, copying code, and any other action that constitutes plagiarism

Final Project

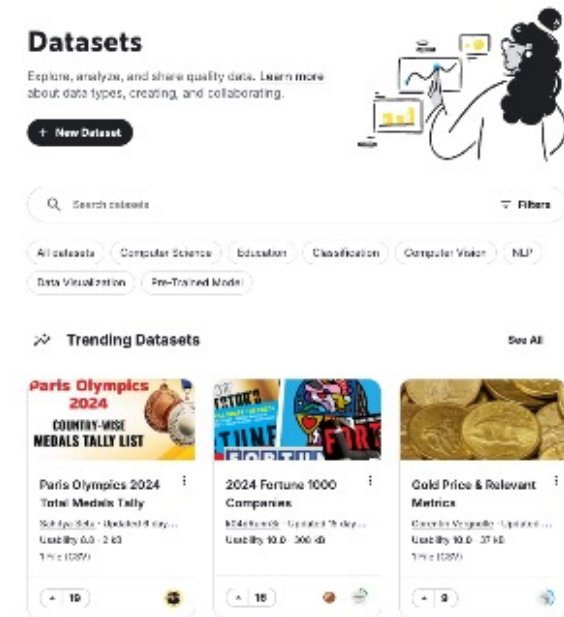
- You will work in (self-formed) teams of 3
- You propose your own topic, gather the data, conduct the analysis, and write your recommendations
- The TA and I are available to provide advice and guidance
- Timeline and deliverables:

<https://fuqua.instructure.com/courses/4169/pages/team-project>

Examples of Potential Project Topics

- Sports
 - Player performance in cricket
 - NBA player salaries
- Entertainment
 - What makes a YouTube videos go viral
 - Which movies will be successful
 - Which restaurants will earn Michelin Stars
- Public Policy
 - Environment, policing, substance abuse, educational outcomes
- Pick topics that are of interest to you!

Dataset Resources



Kaggle Datasets
kaggle.com/datasets

Dataset Search

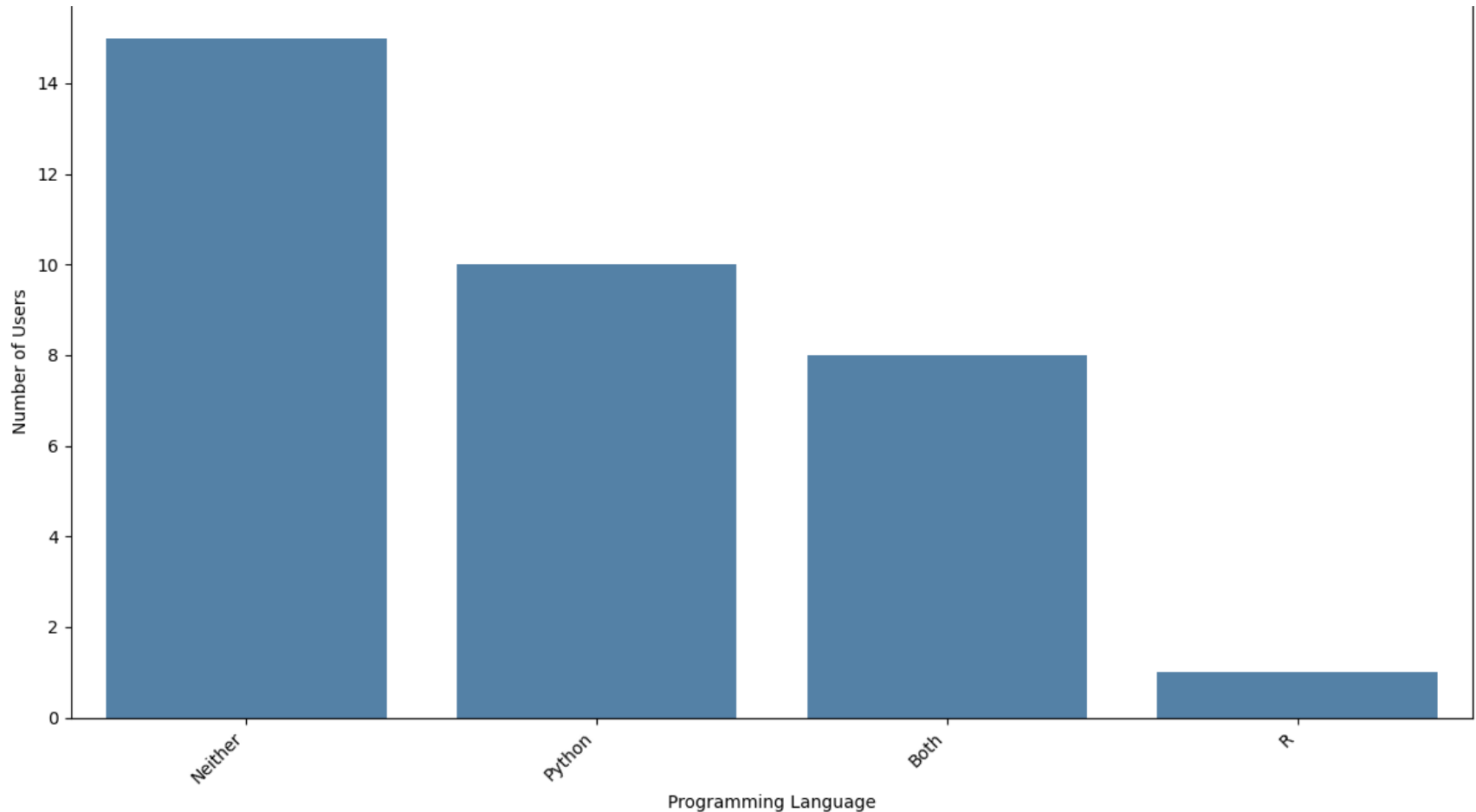


Google Datasets Search
datasetsearch.research.google.com

<https://fuqua.instructure.com/courses/4169/pages/data-sources-for-projects>

Using Python (and a little bit R)

Programming Language used



R and Python

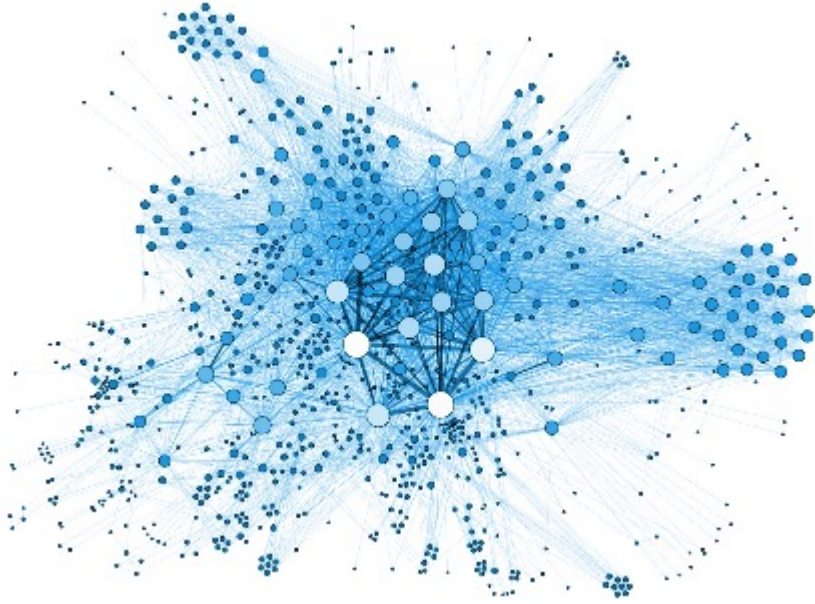
- We will use Python for most of the modeling techniques in the course: linear regression, logistic regression, Random Forests, clustering, text analytics, neural networks, etc.
- We will also work with some R: mostly for CART and collaborative filtering for recommendation systems

R vs. Python

- Both R and Python are popular languages for data analysis
- R is a **statistical** programming language
 - Significantly more powerful than Excel and more user-friendly than other language
 - Can create almost any learning model imaginable
- Python is a **general-purpose** programming language
 - Can do anything
 - Not as simple as R, but still easier to use than many other programming languages

R and Python, continued

- This is not a Python or R programming course
 - The focus is on learning about analytics tools and applications
 - We will use the language; but won't get lost in coding
 - GenAI tools make coding much more accessible
- The scripts we distribute in the course will serve as templates that will cover most tasks you will need to perform



NETWORK ANALYTICS

From Page Rank to Centrality Measures

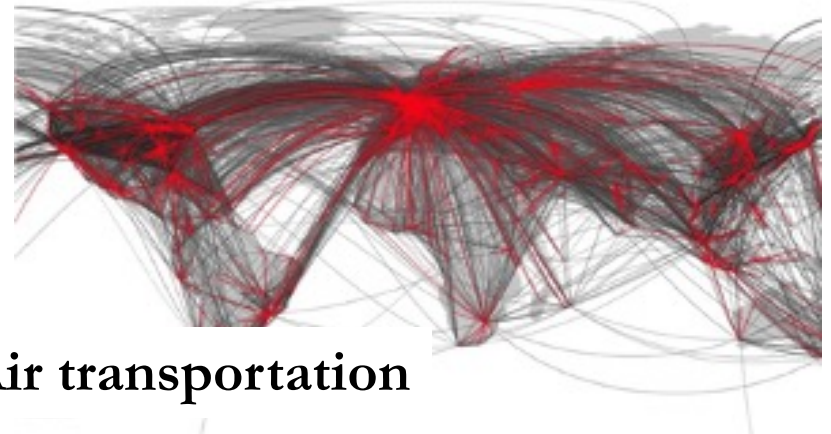
Outline

- Introduction to Networks
 - Applications in business and beyond
 - Notation and terminology
- Centrality measures
 - Measuring the importance of nodes in a network
- Community detection
 - “Clustering” on a network
- Link prediction
 - Predicting which links will be formed in the future

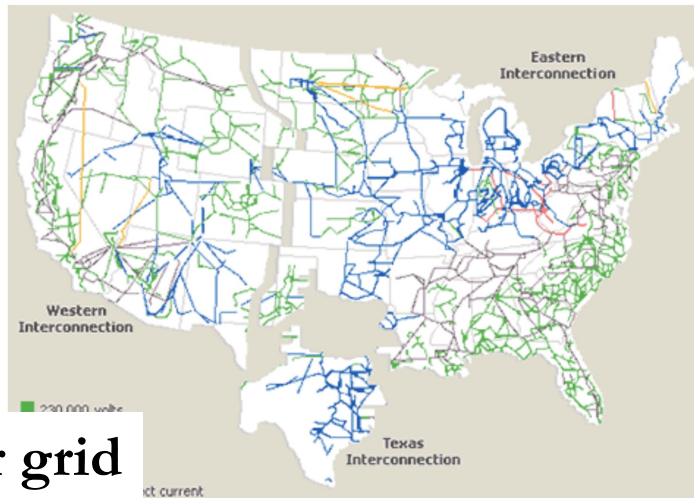
Networks Are Everywhere



Oil pipeline



Air transportation



Power grid



Social network

Networks in Business

- Organization charts
- Collaboration between employees (slack, email)
- Referral graphs (customer, clients)
- Telecommunications (AT&T, Verizon)
- Physical supply chain (i.e., manufacturing)
- Digital supply chain (i.e., software)

Network Lexicon

- **Nodes and Edges**

- A network consists of a collection of nodes (vertices) and connections between nodes called **edges** (arcs, links).
- A, B, C, D, E, F are nodes
- A-F, F-D, etc. are edges

- **Neighbors**

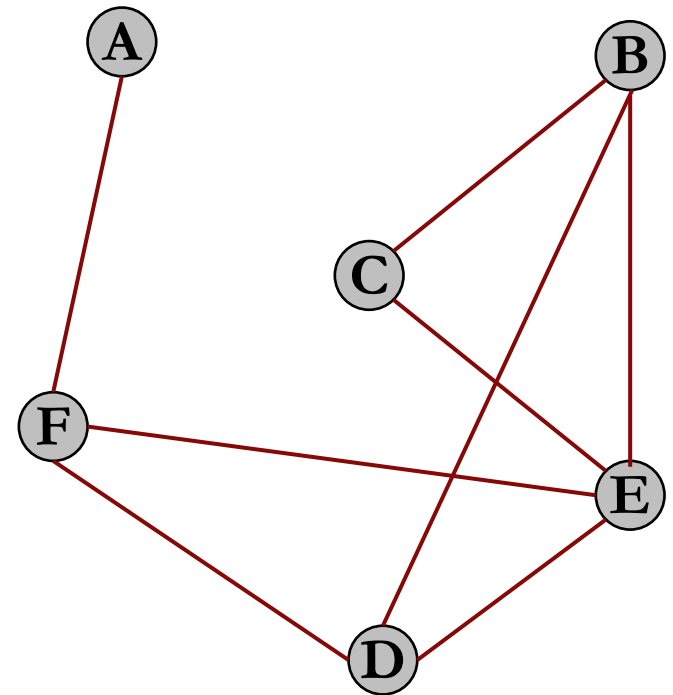
- Node C has two neighbors: B and E

- **Connected**

- Node A is connected to node C: there exists a path from A to C

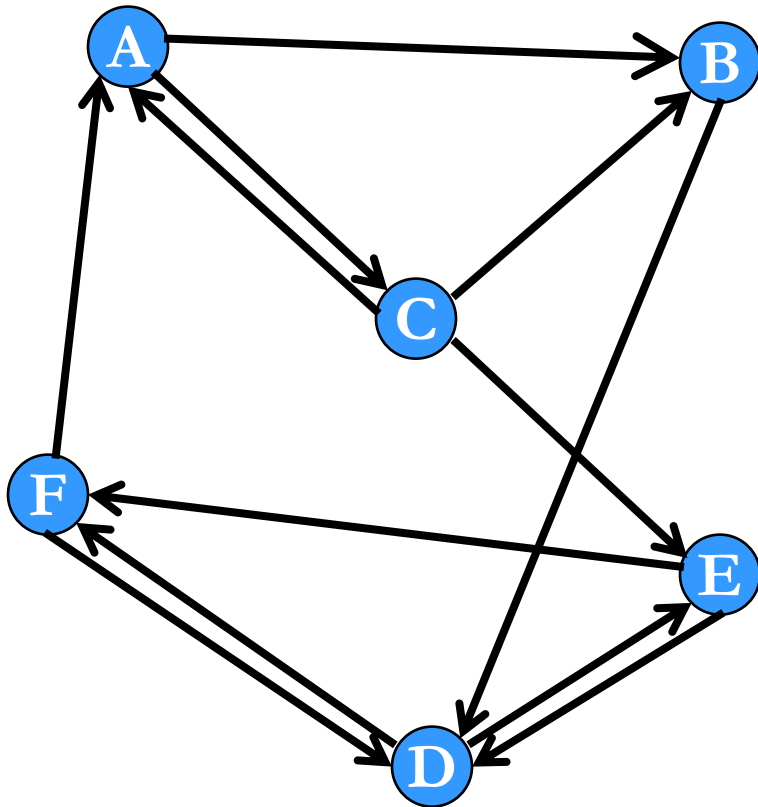
- **Shortest Path and Distance**

- The shortest path from A to C is A-F-E-C
- The distance from A to C is 3 (length of the shortest path)

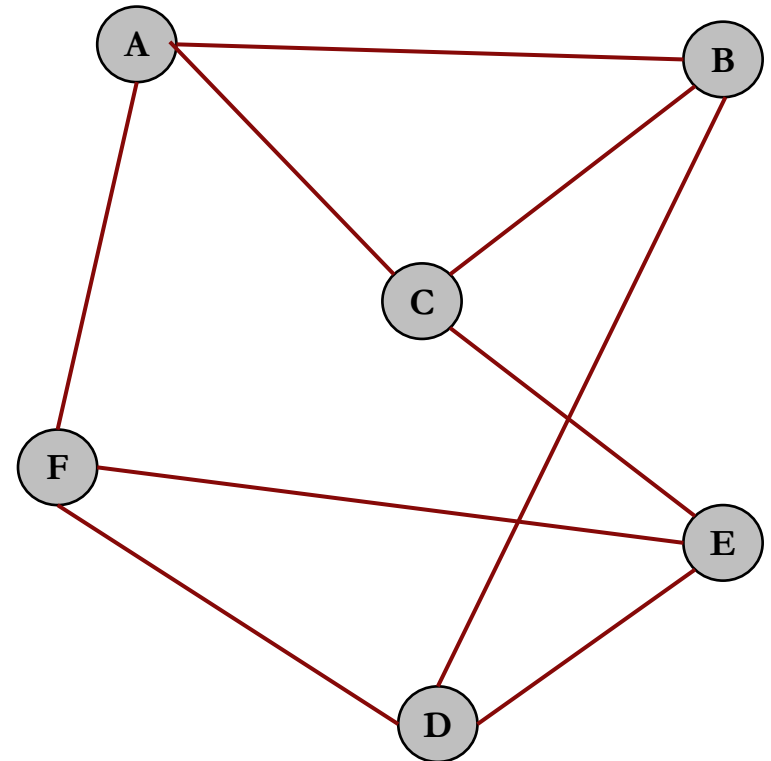


Directed vs. Undirected Networks

Directed Networks
(e.g. Twitter Followers)

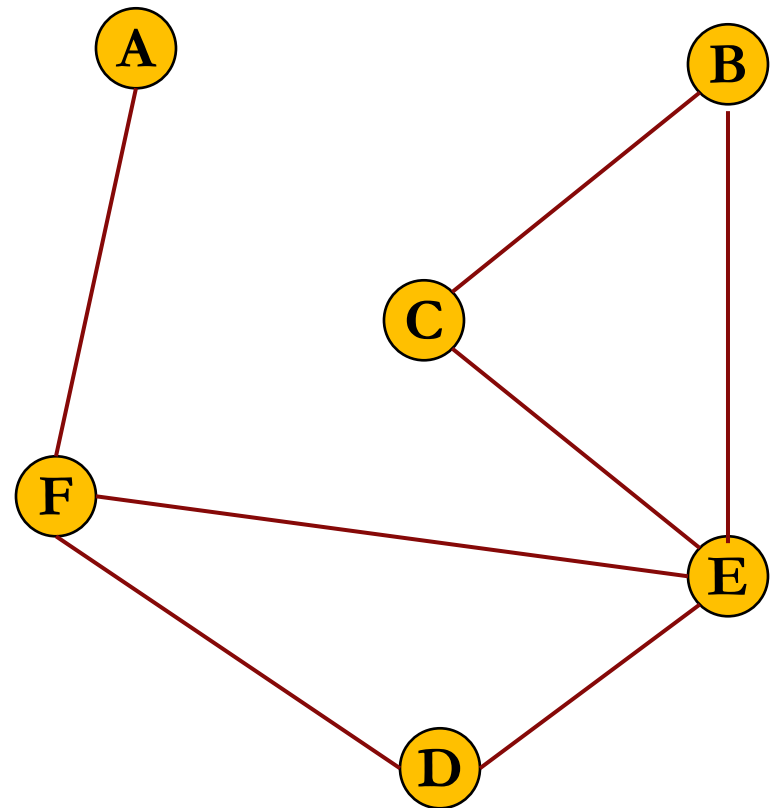


Undirected Networks
(e.g. Facebook Friendships)

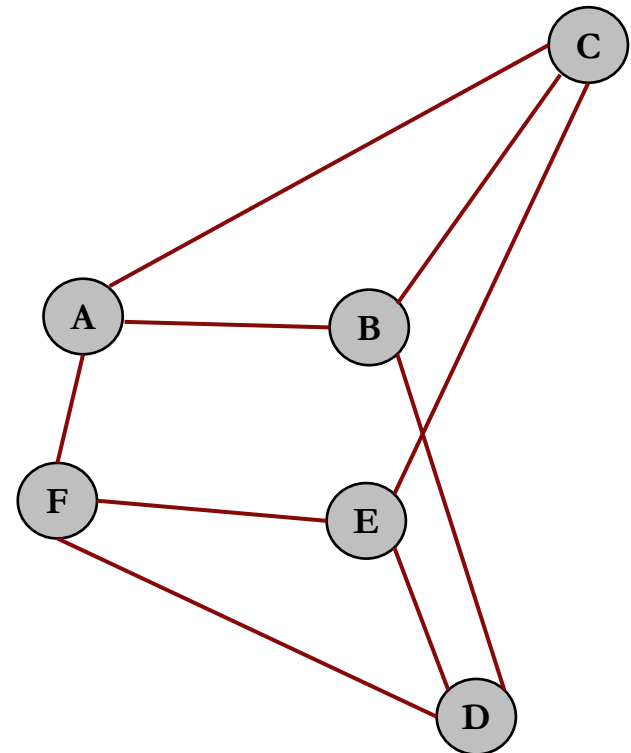
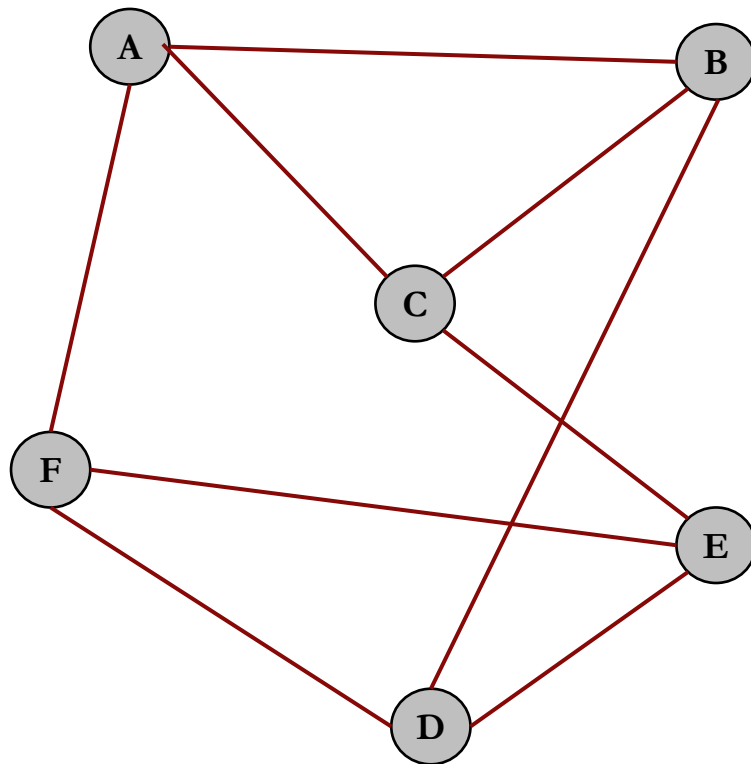


Example: “Market basket” network

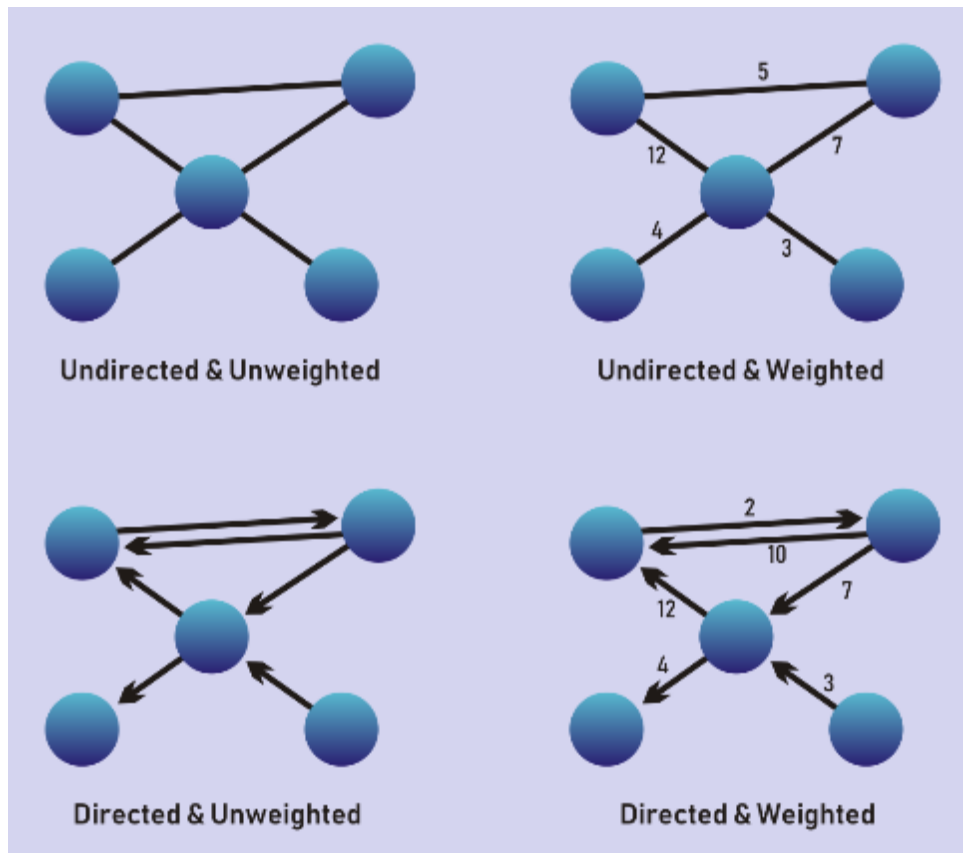
- Each node is a good sold at the supermarket
- Links indicate that the pair of goods are frequently purchased together
- Can be directed or undirected.
- Applications:
 - Designing promotions
 - Understanding customer purchasing patterns
 - Others...



Many Different Ways to Represent the same Network



Weighted vs. Unweighted Networks



- Edges might have weights reflecting importance, distance, cost, capacity, etc.
- This lecture will focus on **undirected, unweighted networks**

Two Running Examples



Emails at Enron

- Energy trading giant from 2001 convicted of fraud
- Federal investigation led to 500,000 internal emails released.

US Flight Network

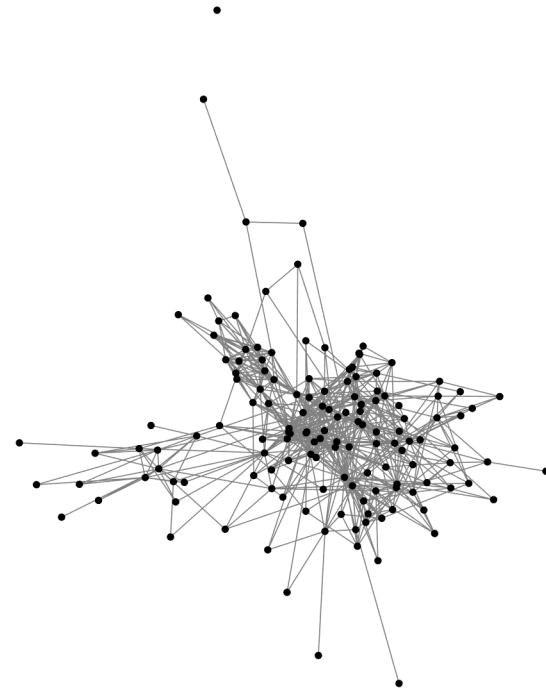
- Publicly available dataset
- Network of 249 airports and 3000+ routes served by commercial airlines

About Enron

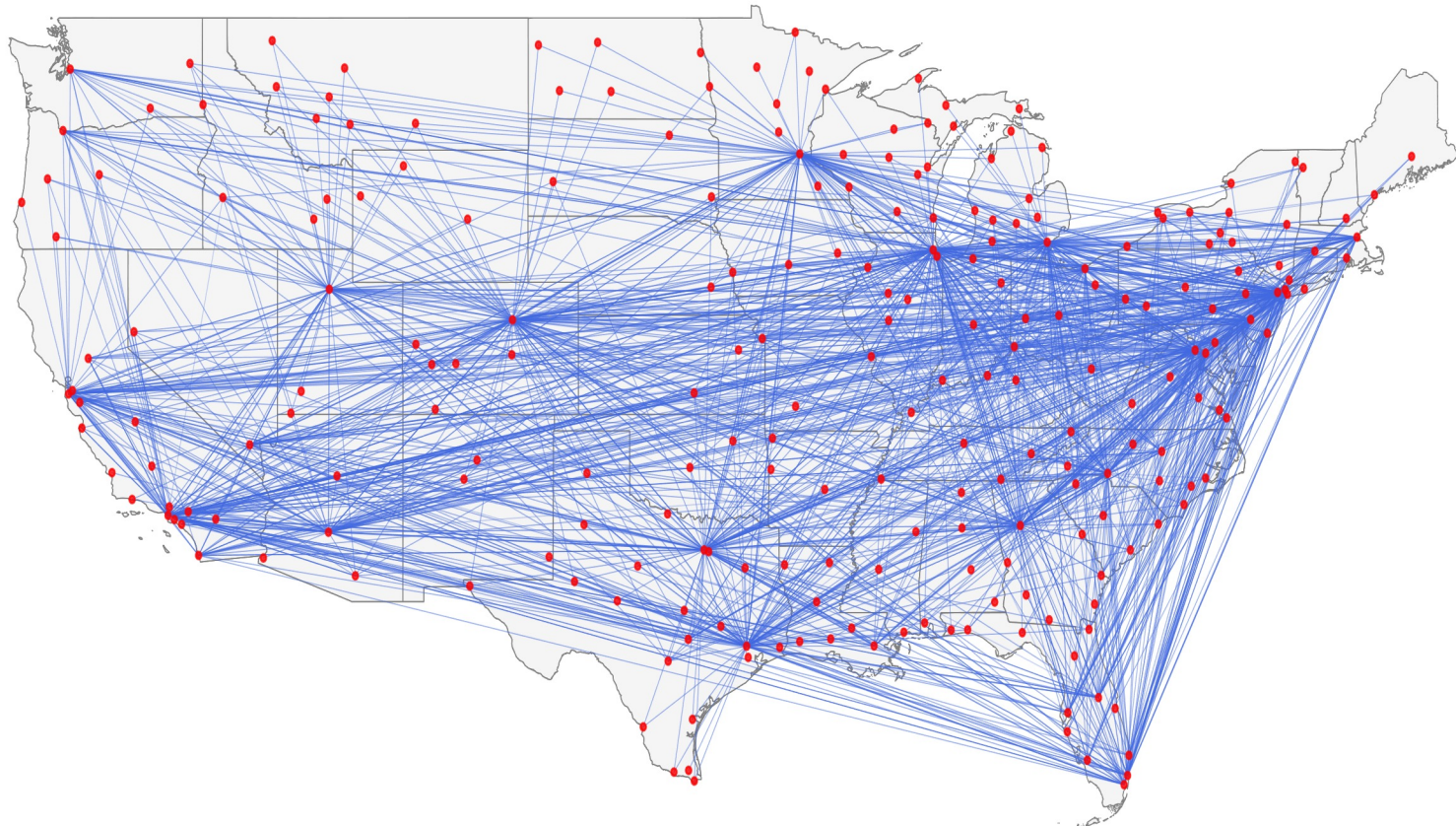
- Enron Corporation: former energy, commodities, and services company based in Houston, TX
 - ~20,000 employees
 - \$100 billion in reported revenues—one of the world's top energy companies
 - Named America's Most Innovative Company from 1996–2001 by *Fortune* magazine
- Implicated in the California energy crisis of 2000-2001
 - Manipulation of electricity prices, illegal shutdown of gas pipelines
- Financial position was found to be sustained by “institutionalized, systematic, and creatively planned accounting fraud”
- Internal emails released by the FERC (Federal Energy Regulatory Commission) as part of their investigation

Enron Email Network

- 148 nodes: former employees
 - 95 energy traders
 - 23 top executives
 - 5 lawyers
 - 25 other employees
 - 678 edges: pairs of employees who exchanged at least 10 emails in ~1999-2000
- Insight on social network of interactions within the company



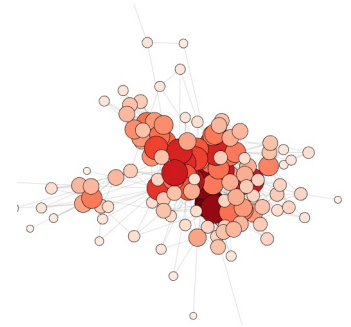
US Air Route Network



Networks: The Analytics Edge

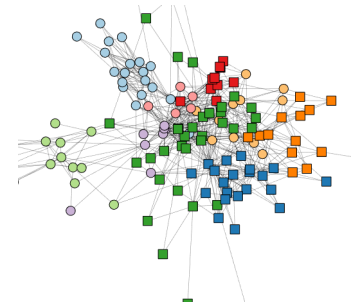
Centrality

- Quantifies the importance / prominence of nodes in a network
 - Several similar, yet different, metrics



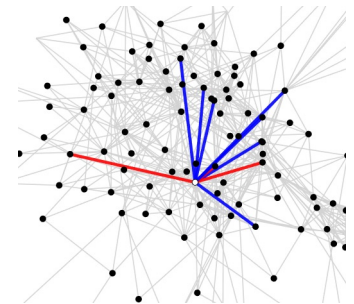
Community detection

- Identifies communities in a network based on existing interactions
 - One among many techniques



Link prediction

- Identifies most “promising” new links to strengthen the network
 - Predictions on *dynamic* networks

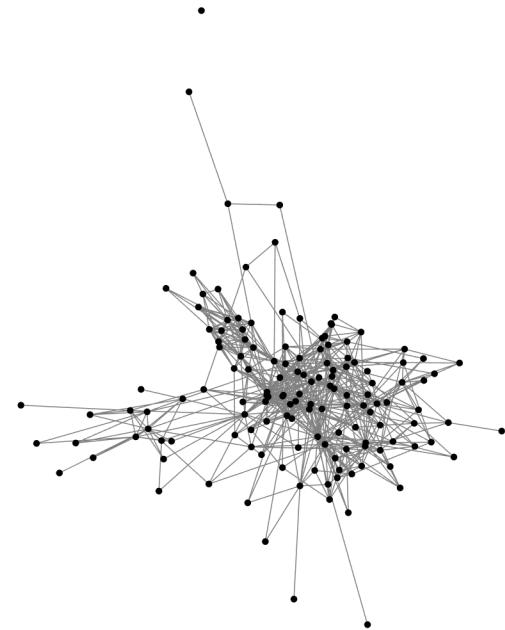


Centrality Measures

Measuring the importance of a node

Network Centrality

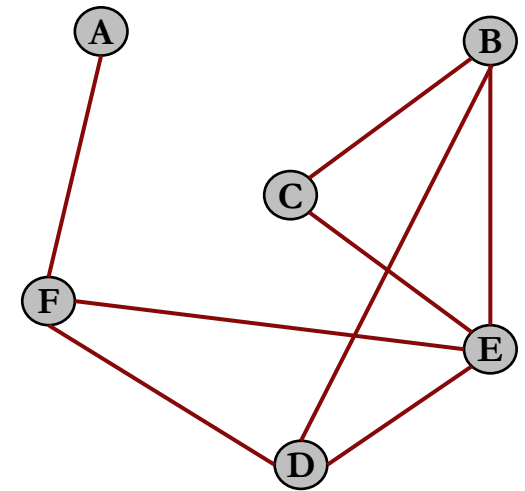
- How central / important are each of the nodes in a network?
- There are many quantitative ways to measure centrality. No metric is perfect.
- We will discuss four centrality metrics
 1. Degree centrality
 2. Closeness centrality
 3. Betweenness centrality
 4. PageRank centrality



Degree Centrality

- **Definition: degree centrality** is the number of neighbors of a node in the network
 - Someone with many LinkedIn connections, many collaborators, etc.
- **Advantage:** simple metric to understand and communicate
- **Limitation:** too local; does not capture connections of those whom you are connected to
 - 10 summer interns count as much as 10 top executives

Node	A	B	C	D	E	F
neighbors	F	C,D,E	B,E	B,E,F	B,C,D,F	A,D,E
Degree centrality	1	3	2	3	4	3

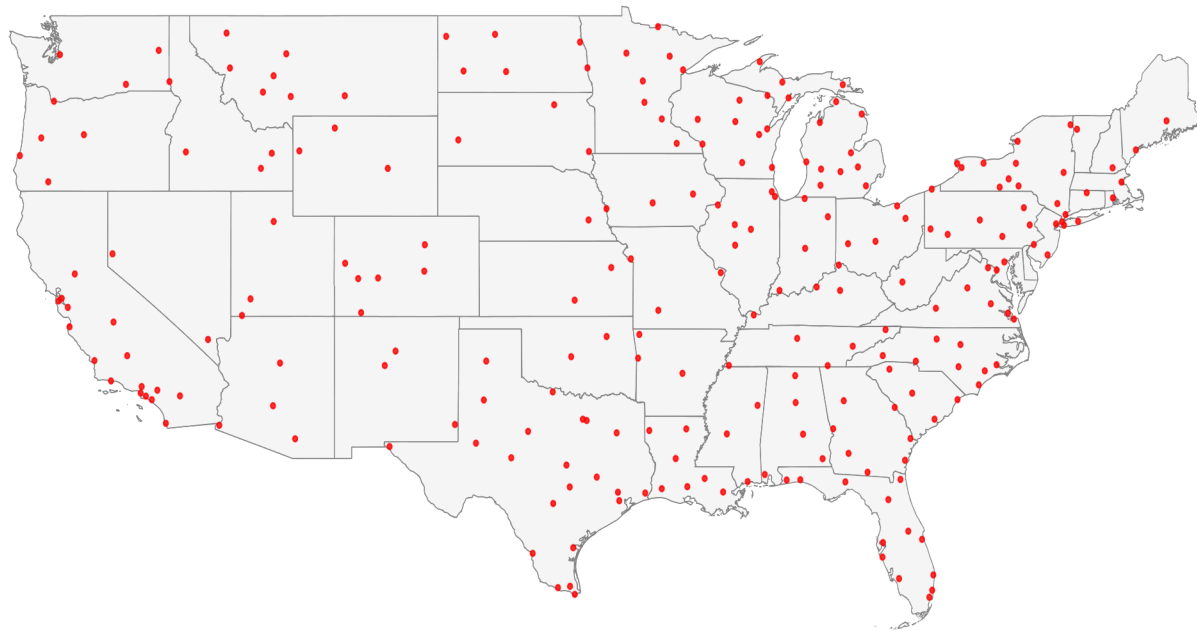


Degree Centrality in Enron Social Network

- Minimum degree of 0
(7 employees are not connected to any others)
- Maximum degree of 31

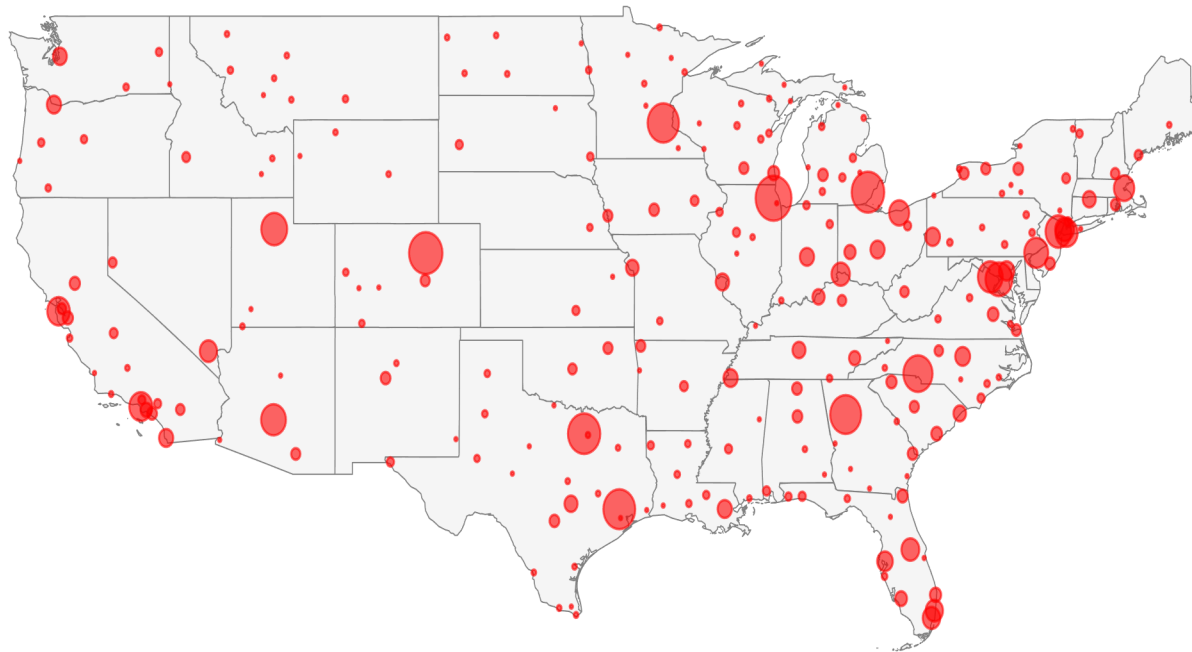


US Airports



Which airports' degree centrality are the highest?

Degree Centrality



Top 5

1. ORD (Chicago): 121
2. DEN (Denver): 105
3. DTW (Detroit): 100
4. DFW (Dallas): 98
5. IAH (Houston): 95

RDU: 21

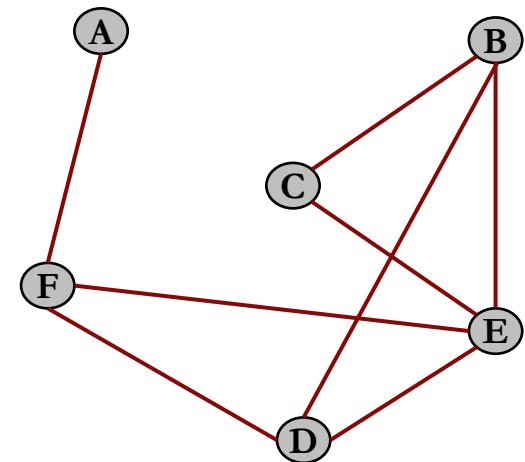
Bottom 5

1. RNT (Renton): 1
2. RST (Rochester): 1
3. LSE (La Crosse): 1
4. JAC (Jackson): 1
5. YUM (Yuma): 1

Closeness Centrality

- **Definition: closeness centrality** is the reciprocal of the sum of the shortest path lengths from that node to every other node. The commonly used **normalized** form multiplies this value by $(n - 1)$, where n is the number of reachable nodes.
 - Someone who needs few introductions to reach everyone on LinkedIn
- **Advantage:** simple way to capture how central each node is
- **Limitation:** too “global”, focused equally on distances to every other node

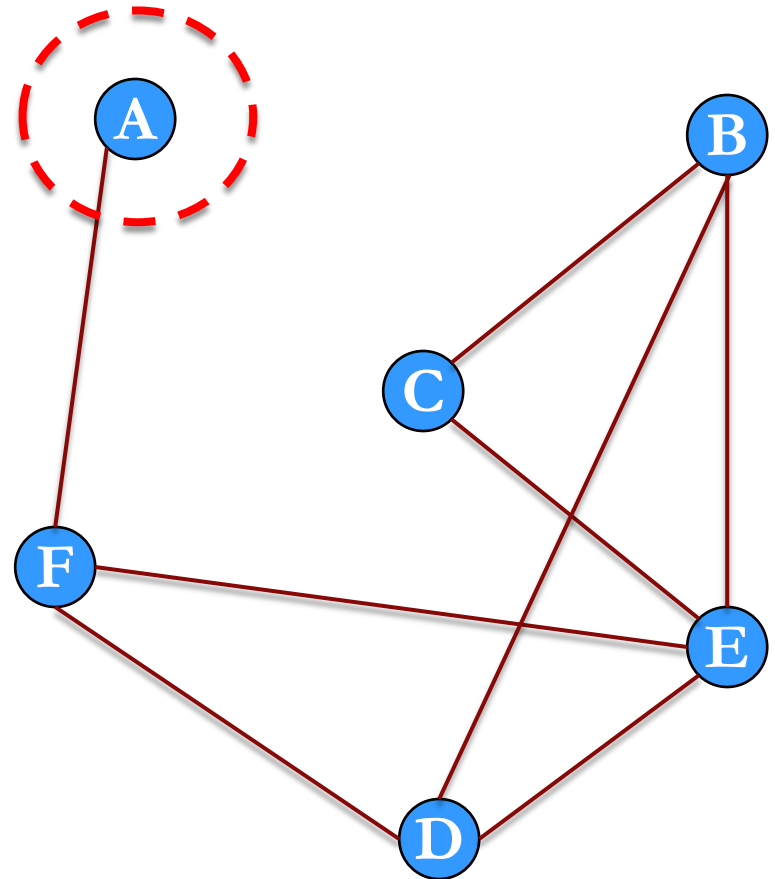
From/To	A	B	C	D	E	F
A	—	3	3	2	2	1
B	3	—	1	1	1	2
C	3	1	—	2	1	2
D	2	1	2	—	1	1
E	2	1	1	1	—	1
F	1	2	2	1	1	—
Sum	11	8	9	7	6	7
Closeness Centrality	5/11	5/8	5/9	5/7	5/6	5/7



Closeness Centrality of Node “A” in the Mininet

Node	Shortest Path	Distance
B	A-F-E-B	3
C	A-F-E-C	3
D	A-F-D	2
E	A-F-E	2
F	A-F	1

- Closeness(A)
= $5 / (3 + 3 + 2 + 2 + 1)$
= $5 / 11 = 0.455$



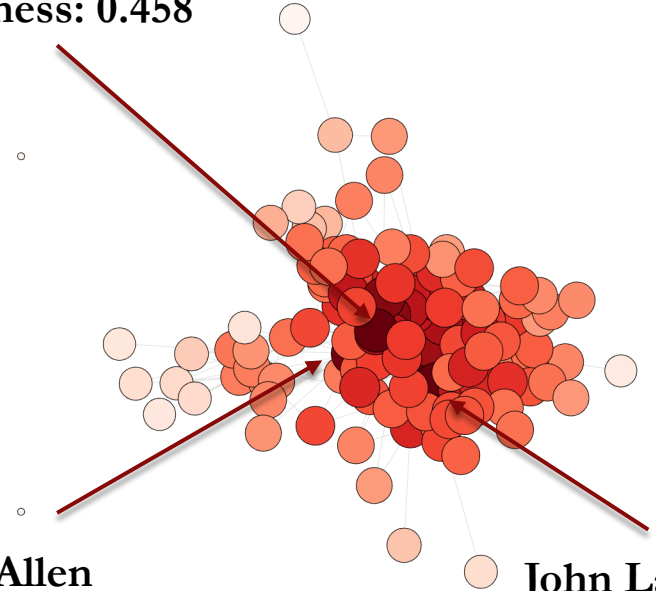
Closeness Centrality in Enron Social Network

- Minimum closeness of 0.214
(average distance of 4.67)
- Maximum closeness of 0.476
(average distance of 2.1)
- Closeness not computed for the
7 nodes that are not in the main
connected component

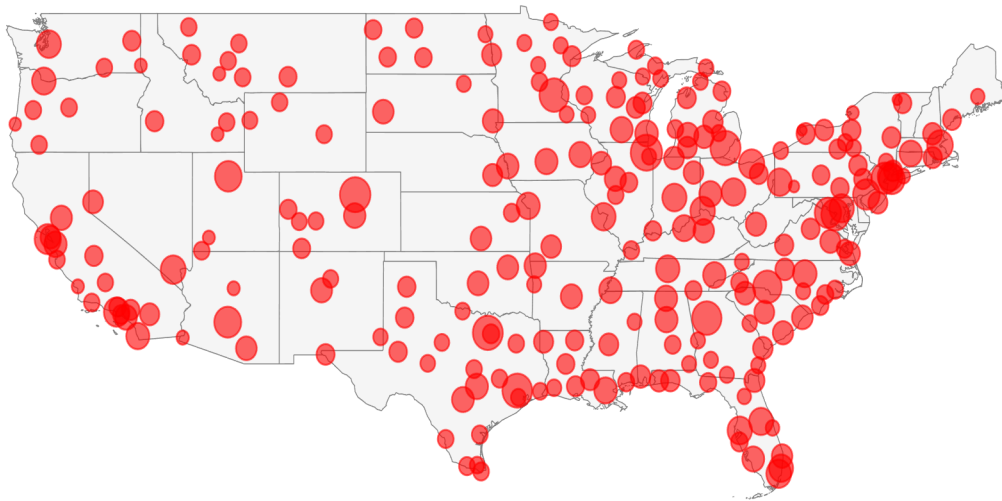
Michael Grigsby
Vice President, ENA
West Region Trading
Closeness: 0.458

Phillip Allen
Managing Director, ENA
West Region Trading
Closeness: 0.471

John Lavorato
CEO, Enron
America
Closeness: 0.476



Closeness Centrality



Top 5

1.	ORD (Chicago):	0.659
2.	DEN (Denver):	0.639
3.	DTW (Detroit):	0.630
4.	DFW (Dallas):	0.627
5.	IAH (Houston):	0.622

RDU: 0.507

Bottom 5

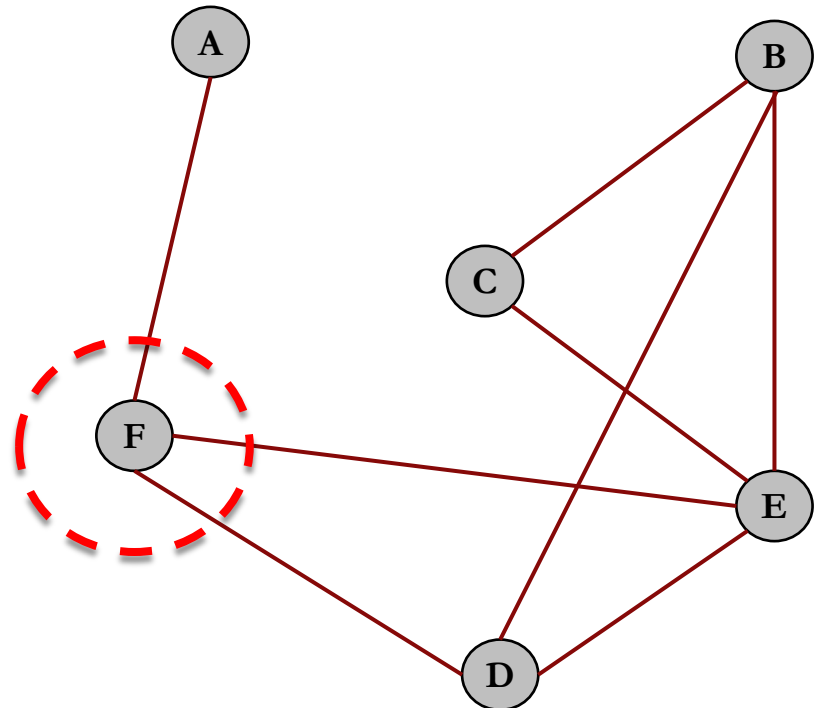
1.	PBG (Plattsburgh):	0.340
2.	IAG (Niagara Falls):	0.340
3.	LBE (Latrobe):	0.347
4.	OTH (North Bend):	0.356
5.	ART (Watertown):	0.359

Betweenness Centrality

- The **betweenness centrality** of a node is the number of shortest paths in the network in which this node appears
- The betweenness of a node v is computed as follows:
 - For every pair of nodes s and t (excluding v), compute the shortest path(s) between s and t , and indicate if v falls along the path
 - If there are multiple “shortest paths” between s and t , indicate instead the proportion of shortest paths that v falls along
 - Sum the result across all pairs of nodes

Betweenness Centrality of Node “F” in the Mininet

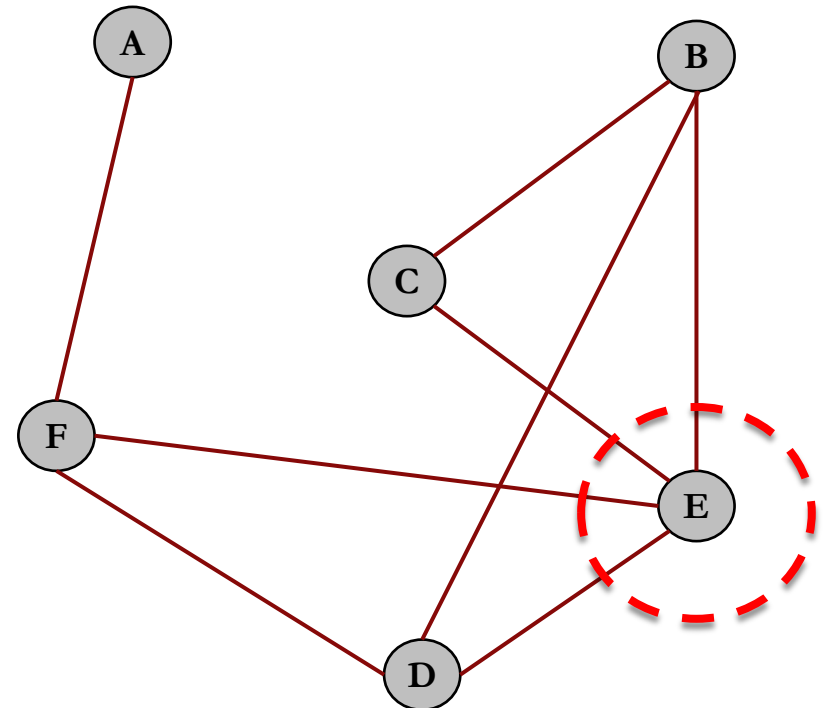
Node Pair	Shortest Path
A-B	A-F-E-B A-F-D-B
A-C	A-F-E-C
A-D	A-F-D
A-E	A-F-E
B-C	B-C
B-D	B-D
B-E	B-E
C-D	C-E-D C-B-D
C-E	C-E
D-E	D-E



- $\text{Betweenness}(F) = 4$

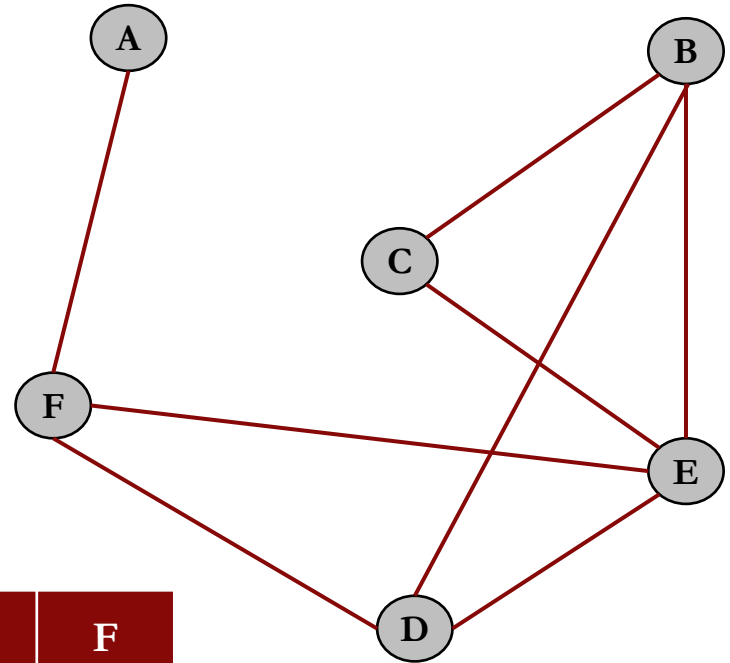
Betweenness Centrality of Node “E” in the Mininet

Node Pair	Shortest Path
A-B	A-F-E-B A-F-D-B
A-C	A-F-E-C
A-D	A-F-D
A-F	A-F
B-C	B-C
B-D	B-D
B-F	B-E-F B-D-F
C-D	C-B-D C-E-D
C-F	C-E-F
D-F	D-F



- $\text{Betweenness}(E) = 3.5$

Betweenness Centrality



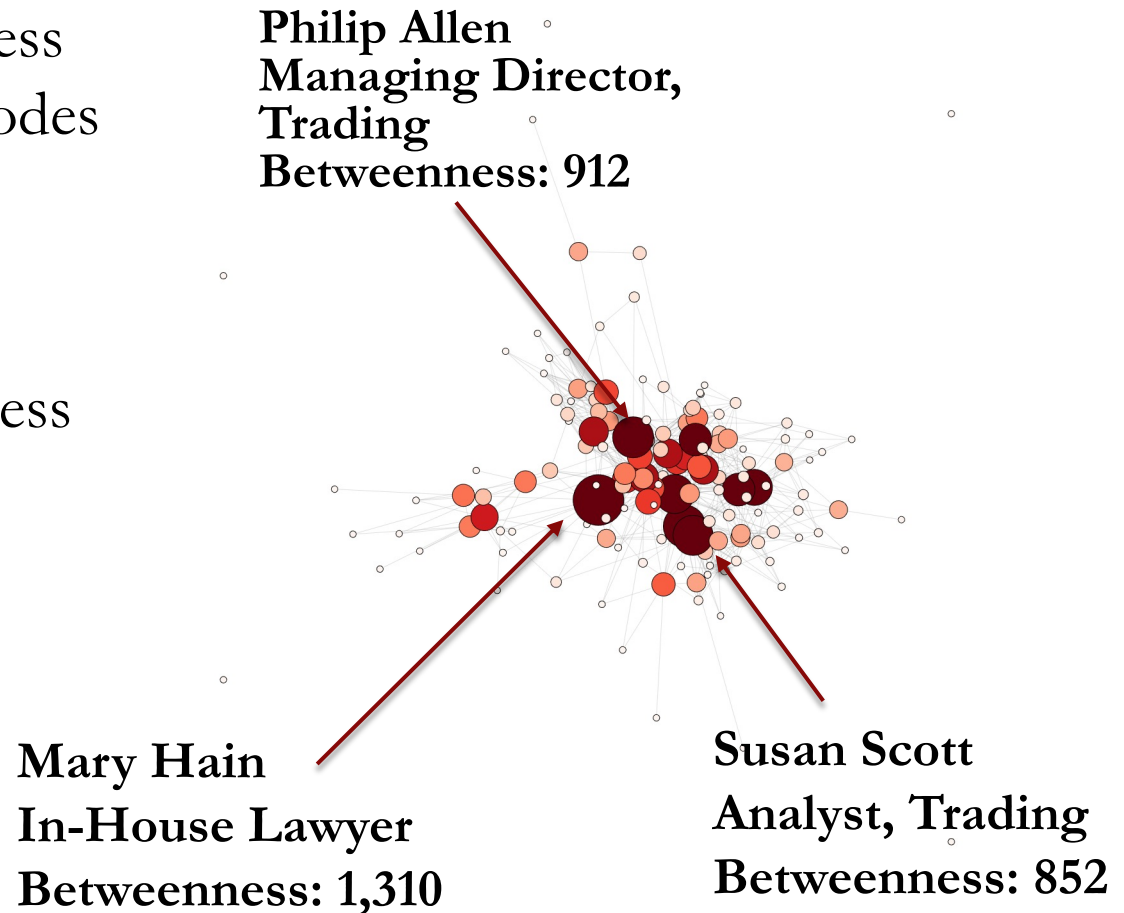
Node	A	B	C	D	E	F
Betweenness centrality	0	0.5	0	1	3.5	4

Betweenness Centrality

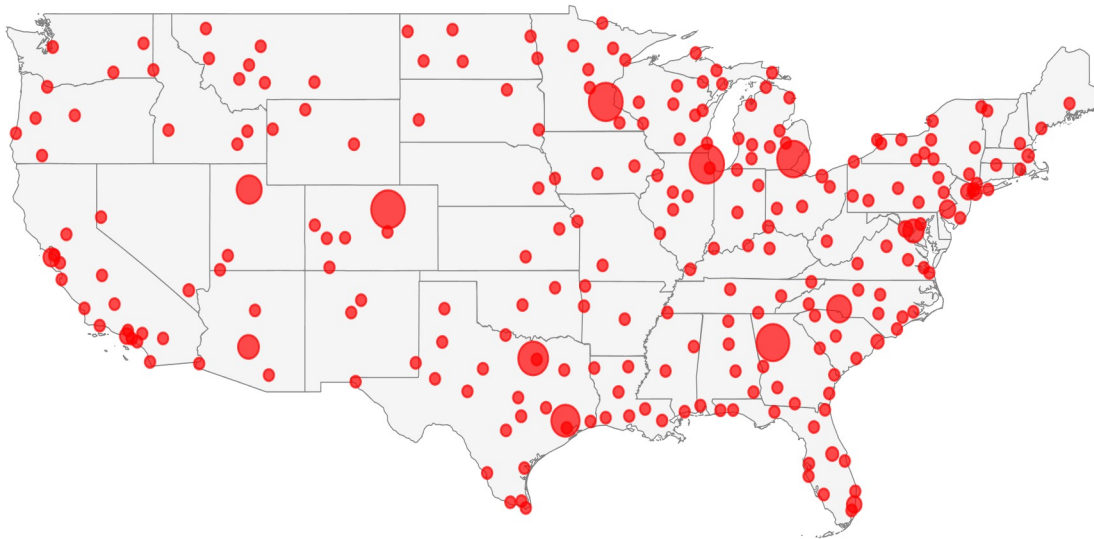
- Betweenness centrality measures the degree to which a node bridges otherwise disconnected parts of the network
 - Example: Somebody who collaborates/communicates with both the data science team and the sales team in the company
 - Someone who “reaches across the aisle” (in language of the United States Congress)
- **Limitation:** Over-emphasizes the shortest paths

Betweenness Centrality in Enron Social Network

- Minimum betweenness centrality of 0 (21 nodes have this value)
- Maximum betweenness centrality of 1,310



Betweenness Centrality



hub-and-spoke model in the US

Top 5

- | | | |
|----|--------------------|-------|
| 1. | ORD (Chicago): | 4,617 |
| 2. | MSP (Minneapolis): | 4,453 |
| 3. | DEN (Denver): | 4,408 |
| 4. | ATL (Atlanta): | 4,153 |
| 5. | DTW (Detroit): | 4,030 |

RDU: 7.83

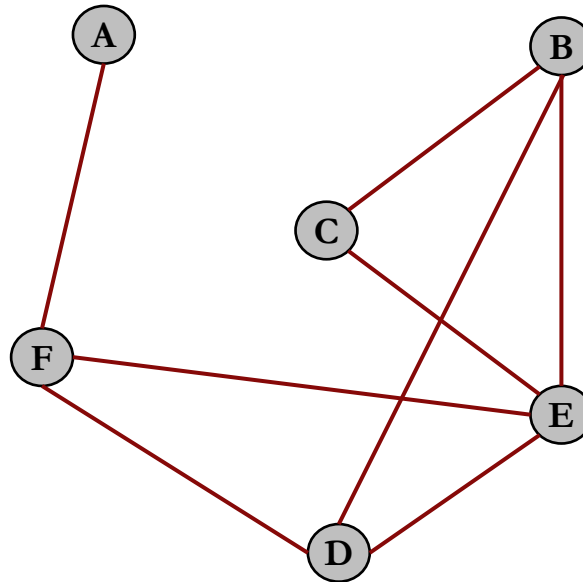
Bottom 5: All 0.00

- | | |
|----|--------------------------|
| 1. | FSM (Fort Smith) |
| 2. | MRY (Monterey Peninsula) |
| 3. | SBA (Santa Barbara) |
| 4. | AVL (Asheville) |
| 5. | FSD (Sioux Falls) |

PageRank

- Recall Google's PageRank algorithm.
- Imagine a randomly “driving” around the network. Which proportion of the time do you visit each node?
 - That is the PageRank centrality (note: sums to 1.00).
- Advantages
 - Robust metric that is hard to “game”
 - Captures intuition that your importance is determined by how many important connections you have.
- Disadvantages
 - Difficult to interpret. What does a PageRank of 0.05 mean?

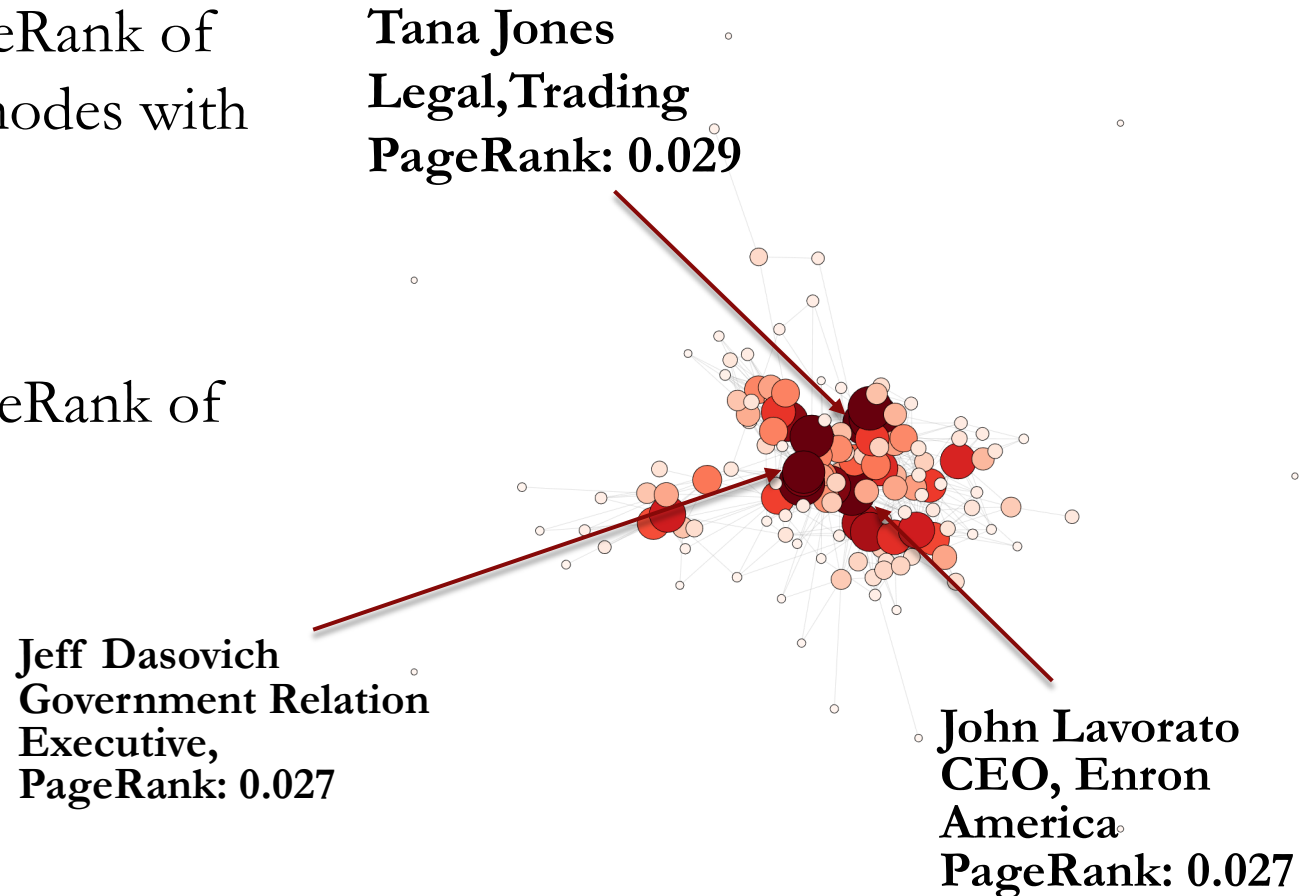
PageRank Metric for the Mininet



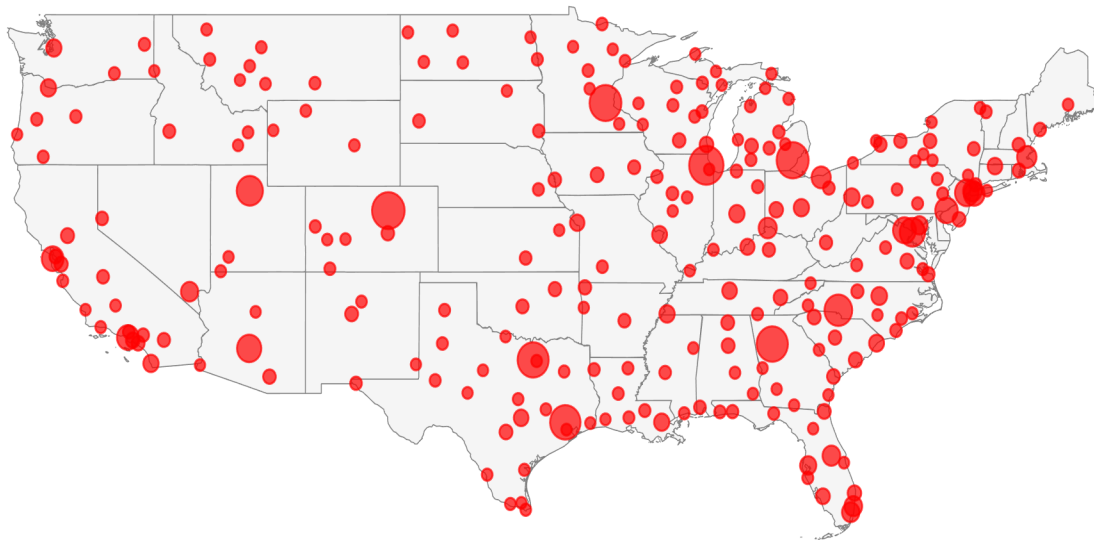
	Node					
	A	B	C	D	E	F
PageRank:	.080	.180	.126	.182	.237	.195

PageRank in the Enron Social Network

- Minimum PageRank of 0.0011 (the 7 nodes with no neighbors)
- Maximum PageRank of 0.029



Page Rank Centrality



Top 5

1. ORD (Chicago): 0.0444
2. DEN (Denver): 0.0395
3. DTW (Detroit): 0.0381
4. MSP (Minneapolis): 0.0376
5. ATL (Atlanta): 0.0360

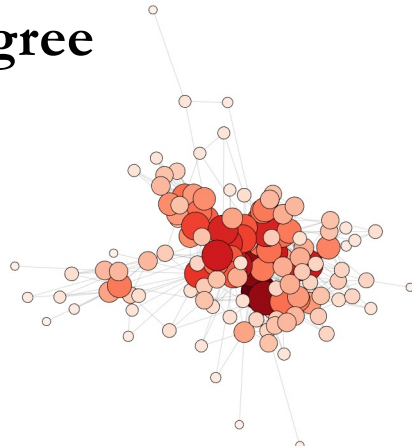
RDU: 0.0067

Bottom 5

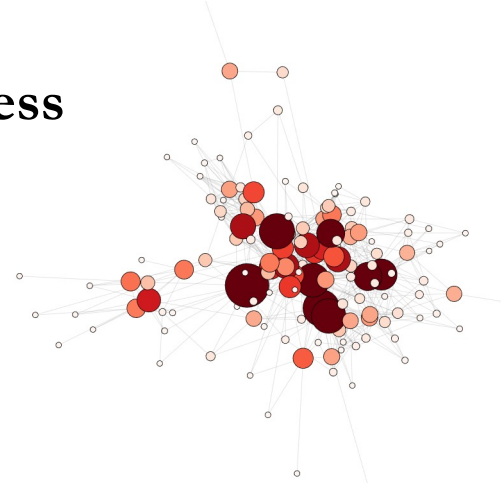
1. ART (Watertown): 0.0008
2. OTH (North Bend): 0.0009
3. ISP (Islip): 0.0009
4. MLB (Melbourne): 0.0009
5. GNV (Gainesville): 0.0009

Centrality in Enron's Network

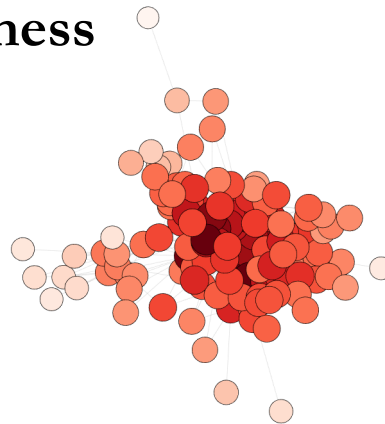
degree



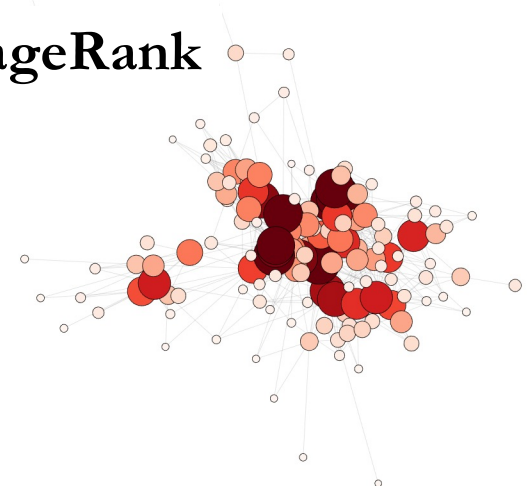
betweenness



closeness



PageRank



Most Central Nodes at Enron

Degree centrality

Rank	Name	Position
1	John Lavorato	President, CEO
2	Philip Allen	Managing Director
3	Michael Grigsby	Vice President

Closeness centrality

Rank	Name	Position
1	John Lavorato	President, CEO
2	Philip Allen	Managing Director
3	Michael Grigsby	Vice President

Betweenness centrality

Rank	Name	Position
1	Mary Hain	In-house lawyer
2	Philip Allen	Managing Director
3	Susan Scott	Analyst - Rotation

PageRank centrality

Rank	Name	Position
1	Tana Jones	Legal
2	Jeff Dasovich	Government Relation Executive
3	John Lavorato	President, CEO

Most Central Airports

Degree Centrality

1. ORD (Chicago)
2. DEN (Denver)
3. DTW (Detroit)
4. DFW (Dallas)
5. IAH (Houston)

Closeness Centrality

1. ORD (Chicago)
2. DEN (Denver)
3. DTW (Detroit)
4. DFW (Dallas)
5. IAH (Houston)

Betweenness Centrality

1. ORD (Chicago)
2. MSP (Minneapolis)
3. DEN (Denver)
4. ATL (Atlanta)
5. DTW (Detroit)

PageRank Centrality

1. ORD (Chicago)
2. DEN (Denver)
3. DTW (Detroit)
4. MSP (Minneapolis)
5. ATL (Atlanta)

Use of Centrality

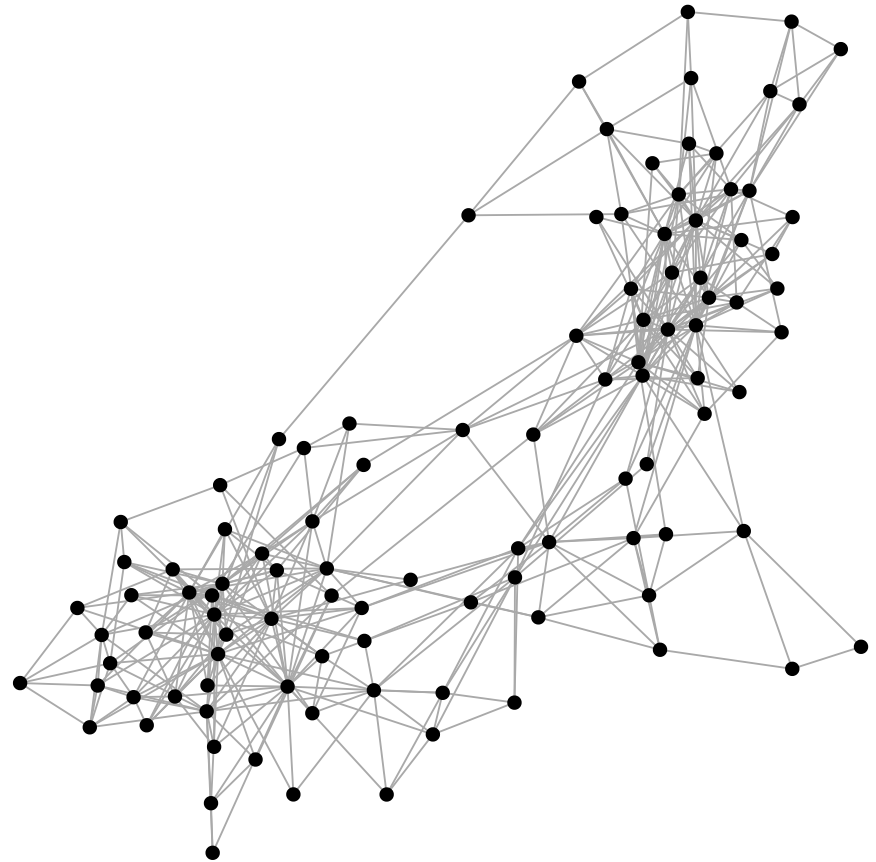
- Central nodes are those that are more connected to the broader network, and can be leveraged for diffusion purposes
 - Organizing a training with a few “central” sales persons and let them diffuse the new sales practices with their peers
 - Providing a new product promotion to a few “central” (or “influential”) customers to encourage word of mouth on a product
- Central nodes can also be leveraged to counter the diffusion of adverse outcomes within an organization
 - Banning “central” accounts to fight fake news propagation
 - In crime prevention, getting a few “central” gang members out of a gang can lead other members to leave the gang

Communities in Social Networks

Finding densely connected nodes

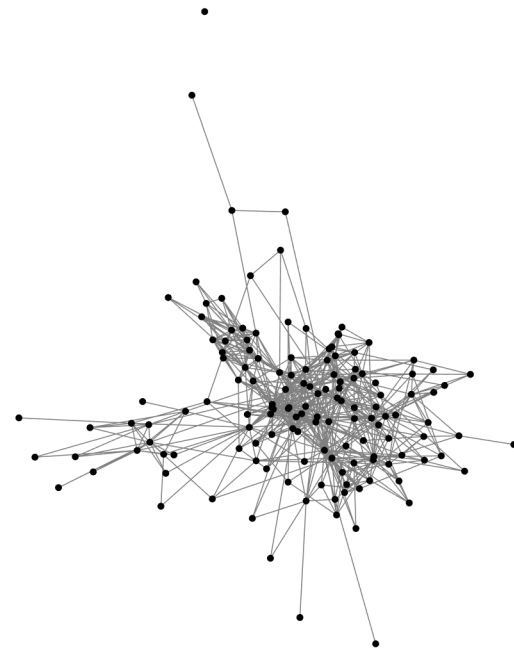
Communities in Networks

- What is a community in a network?
 - Group of nodes that are densely connected within the community versus outside the community.
 - Cluster, Clique
- How might we define communities in the network on the right?



Communities in Social Networks

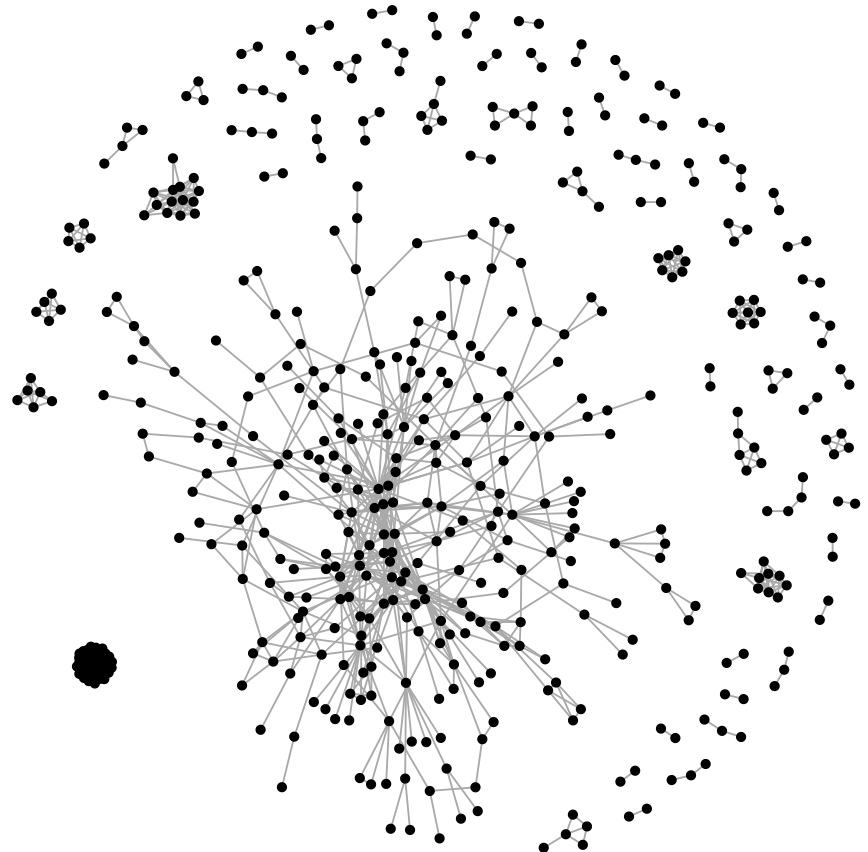
- **Nodes:** People
- **Edges:** Social connections
- **Community:** A group of individuals who are socially connected to each other but not as well connected to others in the network
- **Applications:** Direct marketing, corporate organizational design, ...



Email communication between traders, lawyers, pipeline specialists, and top executives at Enron.

Communities in Word Co-Occurrence Networks

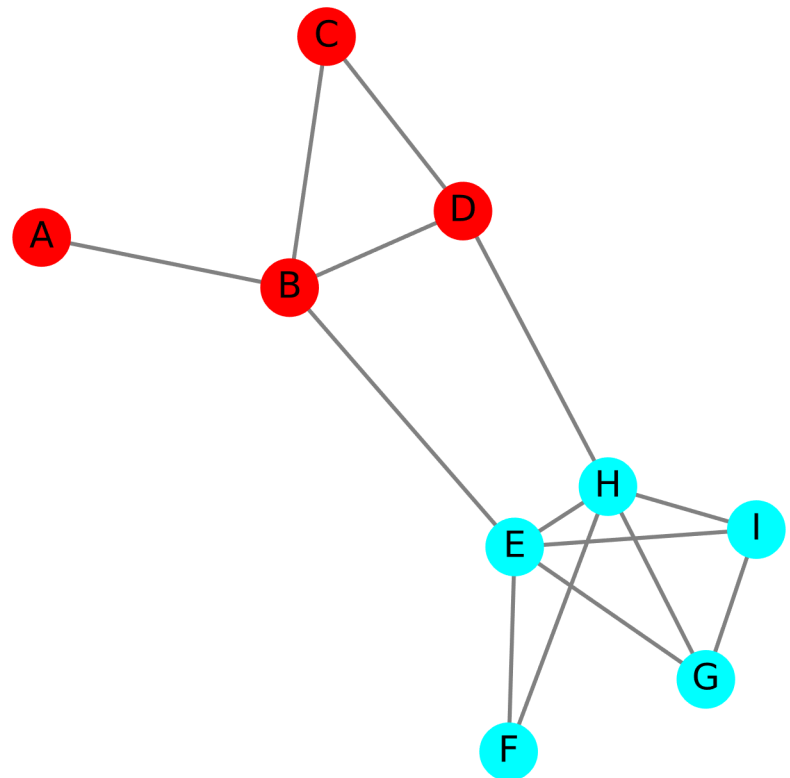
- **Nodes:** Documents (tweets, emails, news articles, ...)
- **Edges:** Documents that share a high number/percentage of word stems in common
- **Community:** Documents that have similar words with each other but different words from other documents
- **Application:** Topic segmentation



Tweets @apple immediately after the iPhone 5c/5s release that shared at least 3 word stems in common.

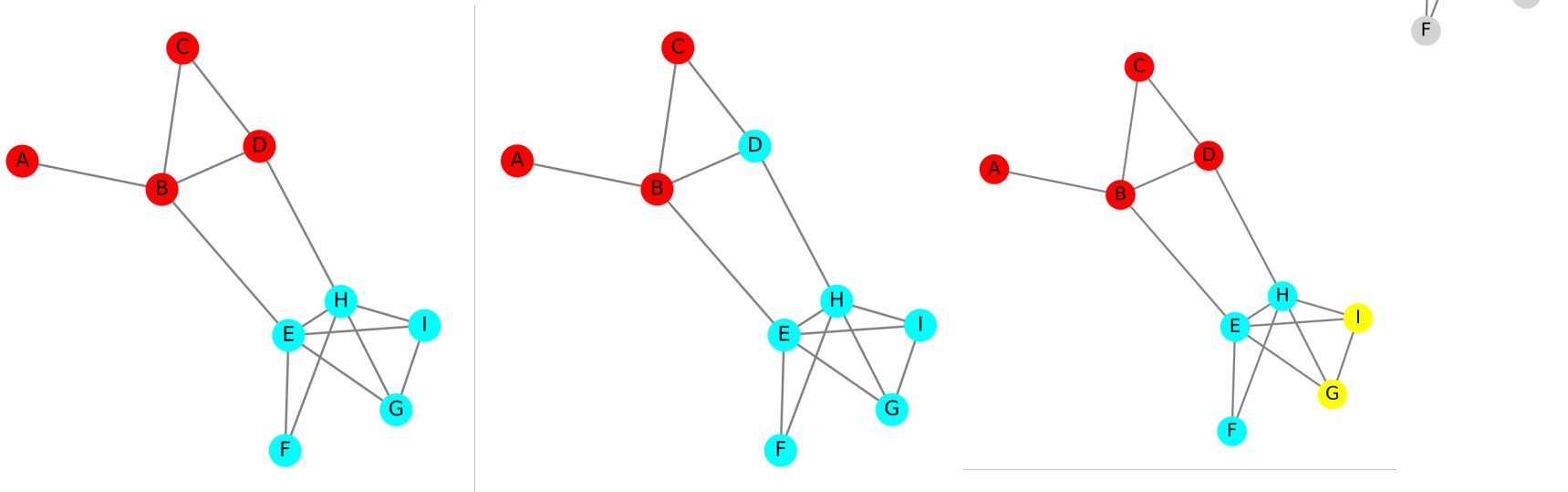
Assessing Community Assignments

- How might we mathematically measure the quality of a community assignment?
- How “good” is the community assignment displayed on the right in this mini-network?
- Is this the “best” community assignment?
- What might “best” mean in this context?



Defining and Assessing Communities

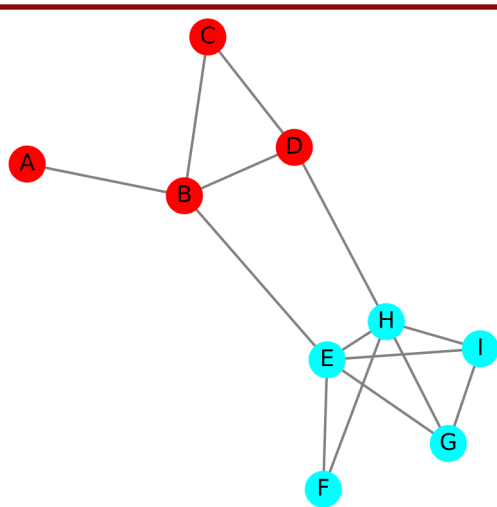
- Can we “detect” communities from network-wide observations?
- How can we evaluate the quality of a community assignment?



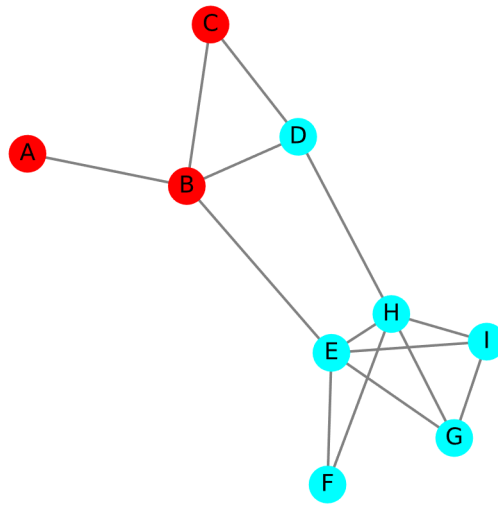
Within-community Links

- **p_{within}** : proportion of edges in the network that link two nodes within the same community
 - How the assignment separates the network into distinct sub-groups

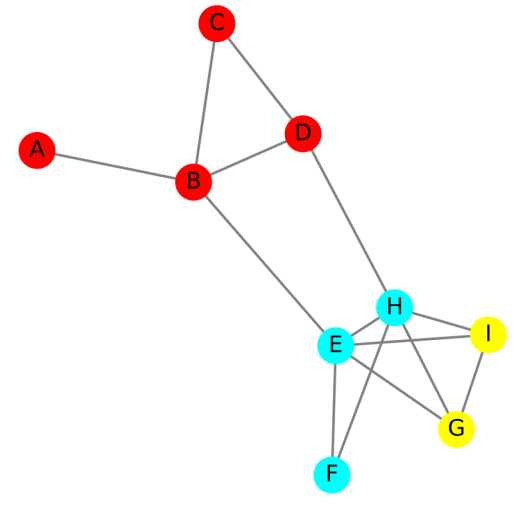
Seems to be “best”. But how to evaluate it?



$$p_{\text{within}} = 12/14 = 0.857$$



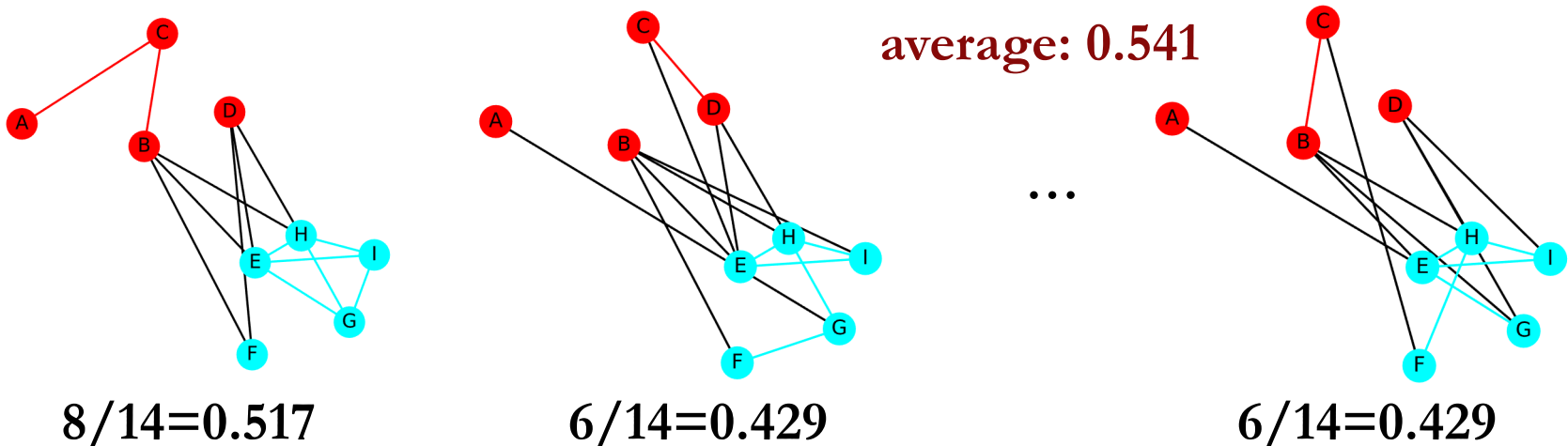
$$p_{\text{within}} = 11/14 = 0.786$$



$$p_{\text{within}} = 8/14 = 0.571$$

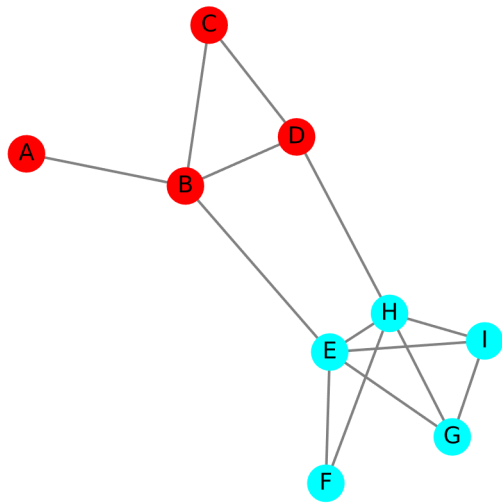
“Baseline” Model and p_{random}

- Consider a collection of “randomized” versions of the network that keep the number of neighbors of each node fixed
- p_{random} : the proportion of edges that link two nodes within the same community, averaged over all the randomized networks
 - proportion of within-community edges that arise by chance

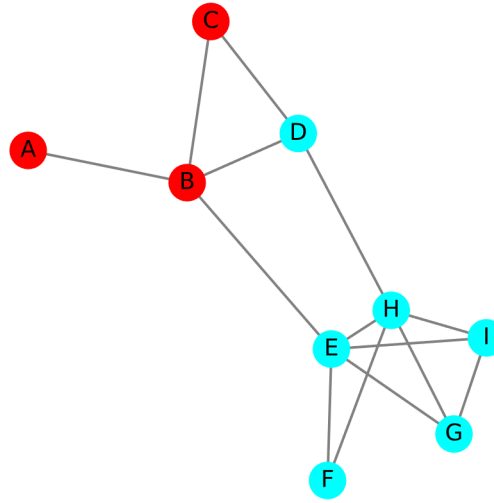


Modularity

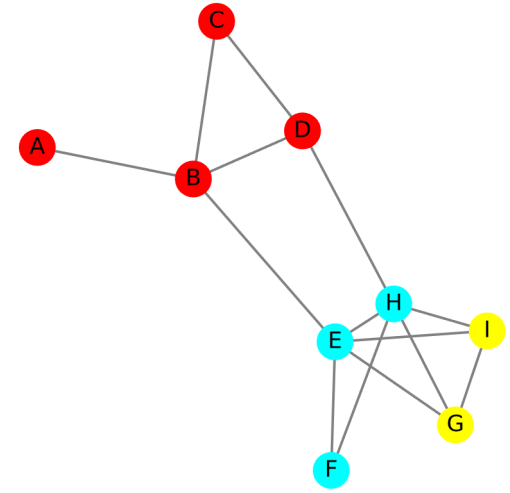
- **Modularity** = $p_{\text{within}} - p_{\text{random}}$ (between -1 and 1)
- Modularity measures the extent to which the within-community links is higher than what would be expected by chance
 - Used to determine the “best” community assignment in the network



Modularity = 0.316



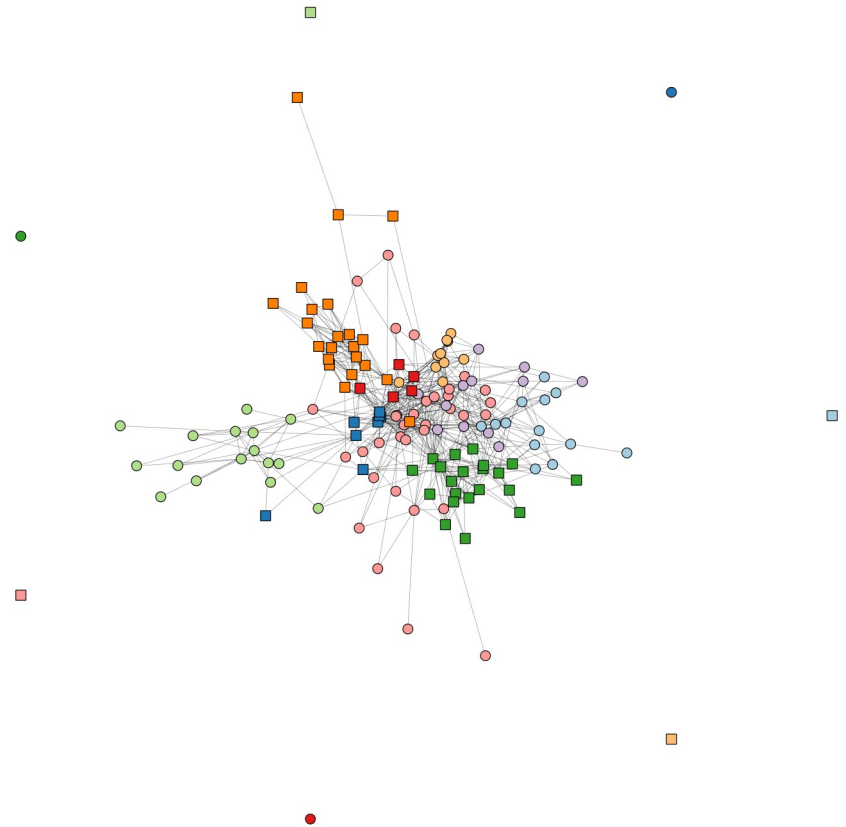
Modularity = 0.161



Modularity = 0.214

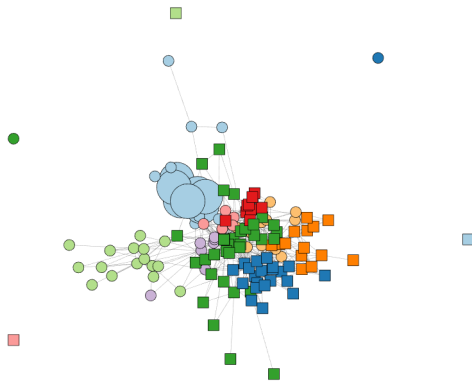
Communities in the Enron Network

- Louvain algorithm aims to find the communities with highest modularity
- 16 communities, ranging from size 1 to size 31
- Modularity is 0.6637
- How do these communities compare to known “**ground truth**” communities in the network?
 - groups based on position
 - groups based on region
 - etc.

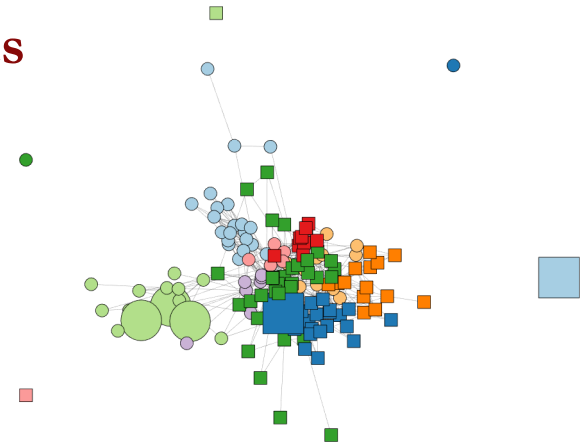


Job Positions in the Enron Network

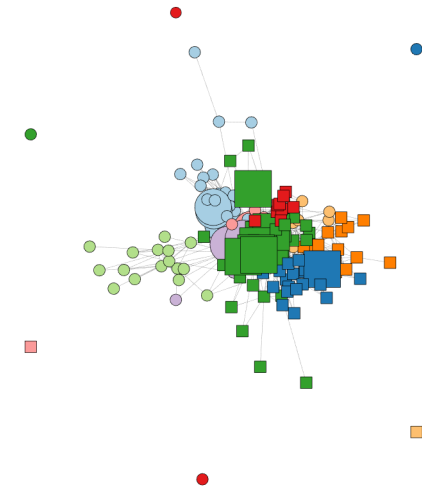
“pipeline”
group



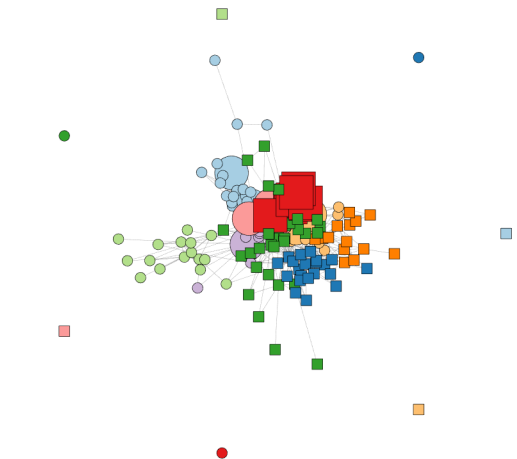
accountants



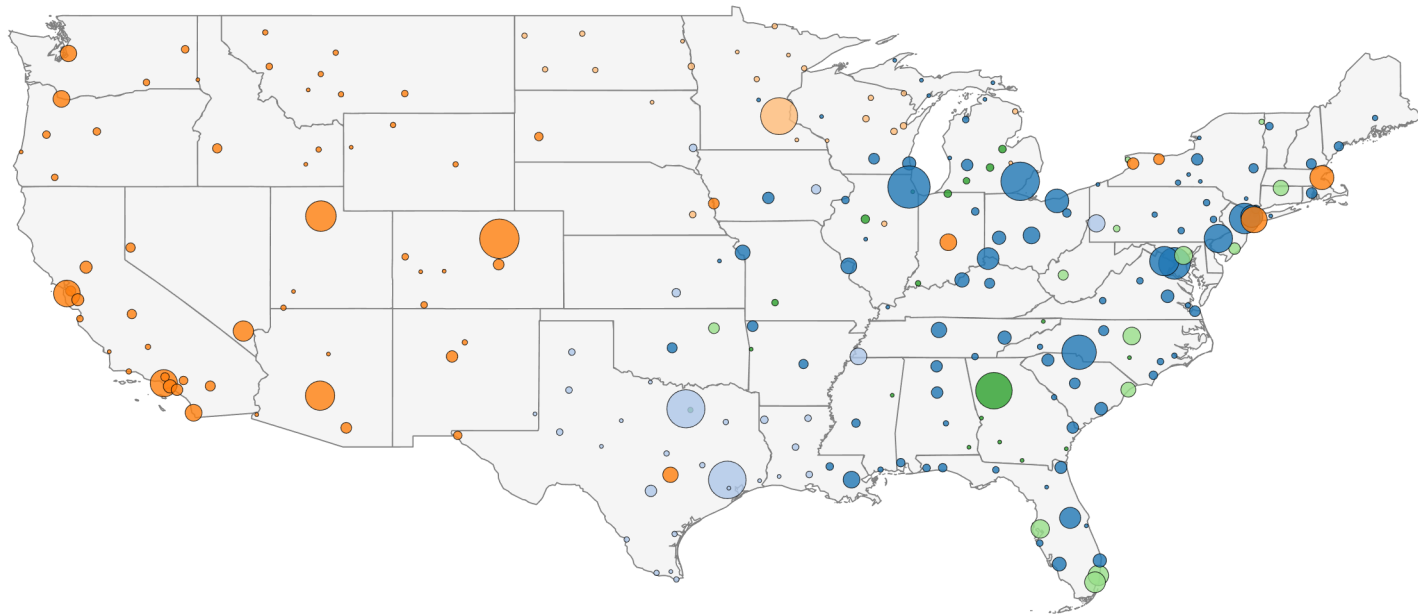
executives



lawyers



Communities in Airline Routes



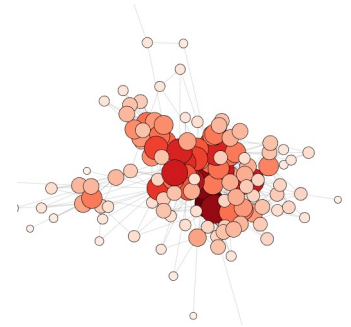
Use of Network Communities

- **Employee networks:** Community detection for (re)organization
 - Differences between the org chart and identified communities can suggest potential organizational realignments within a corporation
 - Business units with multiple poorly connected communities might benefit from interventions to improve communication
- **Consumer networks:** Community detection for customer service
 - Communities can suggest segments and categories of customers, and be used to for targeted actions (e.g., promotions, replies to tweets, etc.)
- **Product networks:** Community detection for product planning
 - Communities can suggest similar products, used for targeted marketing (“other products you might like...”), promotion planning, etc.

Social Networks

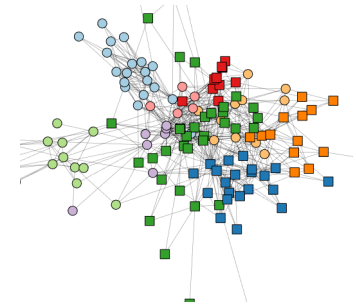
Centrality

- Quantifies the importance / prominence of nodes in a network
 - [Several similar, yet different, metrics]



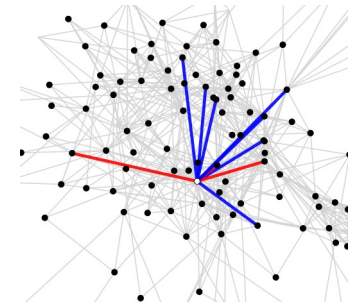
Community detection

- Identifies communities in a network based on existing interactions
 - [One among many techniques]



Link prediction

- Identifies most “promising” new links to strengthen the network
 - Predictions on *dynamic* networks



Link Prediction

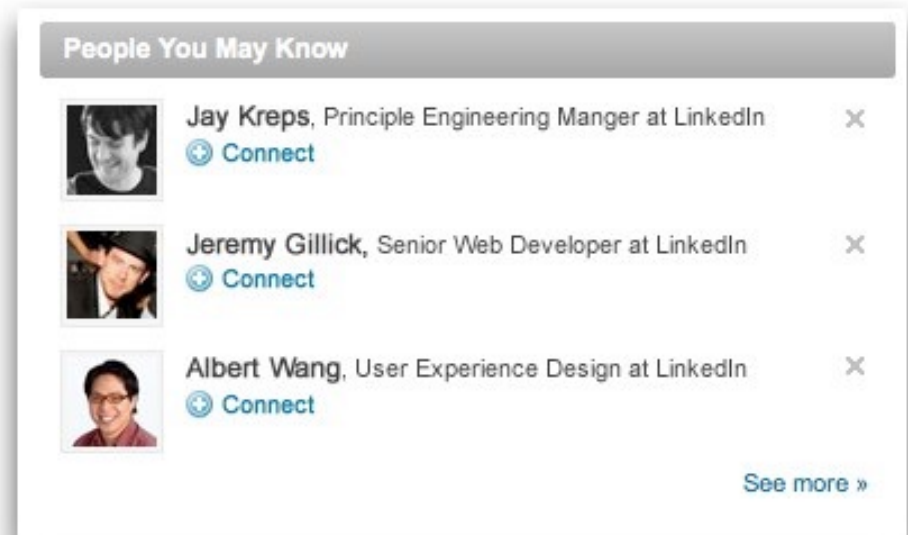
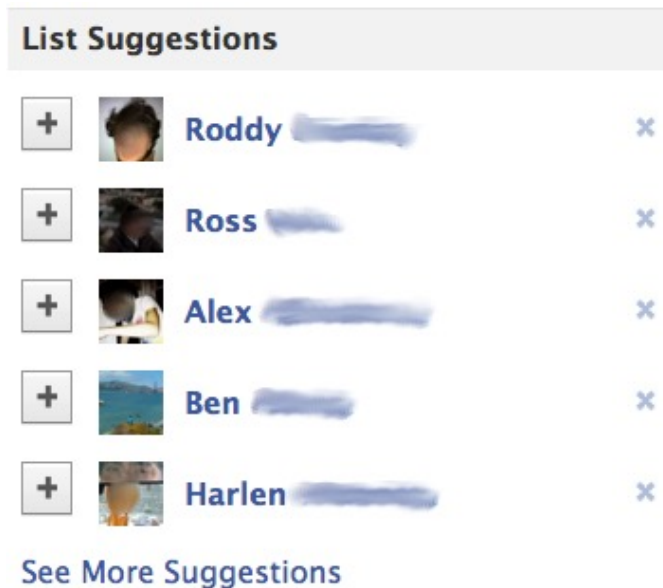
Link Prediction Modeling

- Inputs: Enron network in 2000
 - Node: 139 employees who sent or received an email in 2000
 - Edges: 327 pairs of employees with 10+ emails
- Consideration of the node pairs that were unlinked in 2000
- Dependent variable: whether a pair becomes connected in 2001
 - 276 new edges created in 2001



Friend Recommendations

- Friend recommendations are based on link prediction (who you are likely to know or are likely to be friends with)
- Combination of tools from network analysis and prediction models



Independent Variables

- What variables (available at the end of 2000) can we use to predict whether a link will appear in 2001 if it did not appear in 2000?
 - Only about 3% of all possible unconnected links were linked in 2001
- Network information (from the 2000 network)
 - For each unlinked pair of 2 nodes in 2000, the shortest-path distance between them
 - Minimum/maximum PageRank, betweenness, closeness, and degree centrality of 2 nodes (**PageRankMin**, **PageRankMax**, **BetweennessMin**, etc.)
 - Proportion of neighbors in common
 - Whether the two nodes are in the same community
- Information about the employees themselves
 - Whether the nodes are in the same functional area (traders, lawyers, execs, etc.)
- **k : number of independent variables ($k = 13$)**
- **n : number of observations ($n = 9,264$)**

Link Prediction Data

k : # of independent variables ($k = 13$); n : # of observations ($n = 9,264$)

	FutureLink	Connected	Distance	PageRankMin	PageRankMax	...	NeighborsInCommon	CommunityMatch	AreaMatch
0	0	1	3	0.001684	0.006443	...	0.0	1	1
1	0	1	3	0.004953	0.006443	...	0.0	0	1
2	0	1	3	0.002867	0.006443	...	0.0	0	1
3	0	1	3	0.002008	0.006443	...	0.0	0	1
4	0	1	2	0.002317	0.006443	...	0.333333	1	1
5	0	1	6	0.006443	0.010327	...	0.0	0	0
6	0	1	3	0.006443	0.019754	...	0.0	0	0
7	0	1	4	0.001835	0.006443	...	0.0	0	1
8	1	1	2	0.006443	0.021382	...	0.111111	1	1
9	0	1	5	0.001627	0.006443	...	0.0	0	0
10	0	1	3	0.005253	0.006443	...	0.0	0	1
11	0	1	3	0.00268	0.006443	...	0.0	0	0
12	0	1	3	0.006443	0.00899	...	0.0	0	0
13	0	0	0	0.001283	0.006443	...	0.0	0	0
14	0	1	5	0.003025	0.006443	...	0.0	0	0
15	0	0	0	0.001283	0.006443	...	0.0	0	0
...	...	...	...	...	...	...	...	...	...
9259	0	0	0	0.001283	0.003263	...	0.0	0	0
9260	0	0	0	0.001283	0.003263	...	0.0	0	0
9261	0	0	0	0.001283	0.015594	...	0.0	0	0
9262	0	0	0	0.001283	0.015594	...	0.0	0	0
9263	0	0	0	0.001283	0.001283	...	0.0	0	1

Training and Testing Set (70% Random Split)

Full dataset ($n = 9,264$)

	FutureLink	...	CommunityMatch	AreaMatch
0	0	...	1	1
1	0	...	0	1
2	0	...	0	1
3	0	...	0	1
4	0	...	1	1
5	0	...	0	0
6	0	...	0	0
7	0	...	0	1
8	1	...	1	1
9	0	...	0	0
10	0	...	0	1
11	0	...	0	0
12	0	...	0	0
13	0	...	0	0
14	0	...	0	0
15	0	...	0	0
...	...	...	...	...
9262	0	...	0	0
9263	0	...	0	1

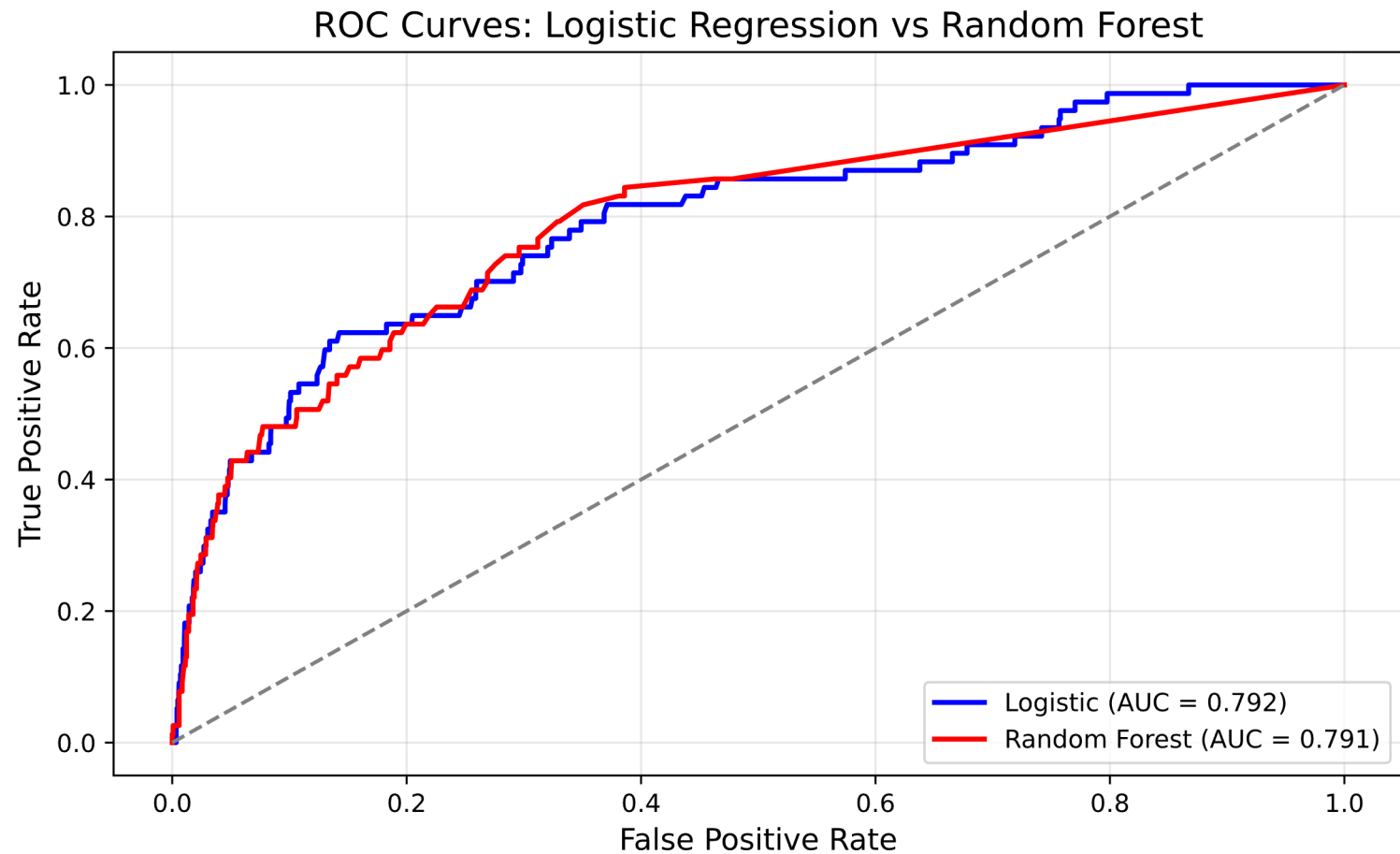
Training data ($n = 6,484$)

	FutureLink	...	CommunityMatch	AreaMatch
7150	0	...	0	1
6573	0	...	0	0
1176	0	...	0	0
5268	0	...	0	0
7318	0	...	0	0
7347	0	...	0	0
2301	0	...	0	0
6134	0	...	0	1
...	...	...	...	...
7647	0	...	0	0
7071	0	...	0	0

Test data ($n = 2,780$)

	FutureLink	...	CommunityMatch	AreaMatch
5880	0	...	1	1
2779	0	...	0	1
7896	0	...	1	1
3460	0	...	0	0
4271	0	...	0	1
8302	0	...	0	0
4676	0	...	0	1
488	0	...	0	0
...	...	...	...	...
2664	0	...	0	1
6725	0	...	1	1

Test-Set ROC Curve



Logistic Regression Output

Logistic Regression Results

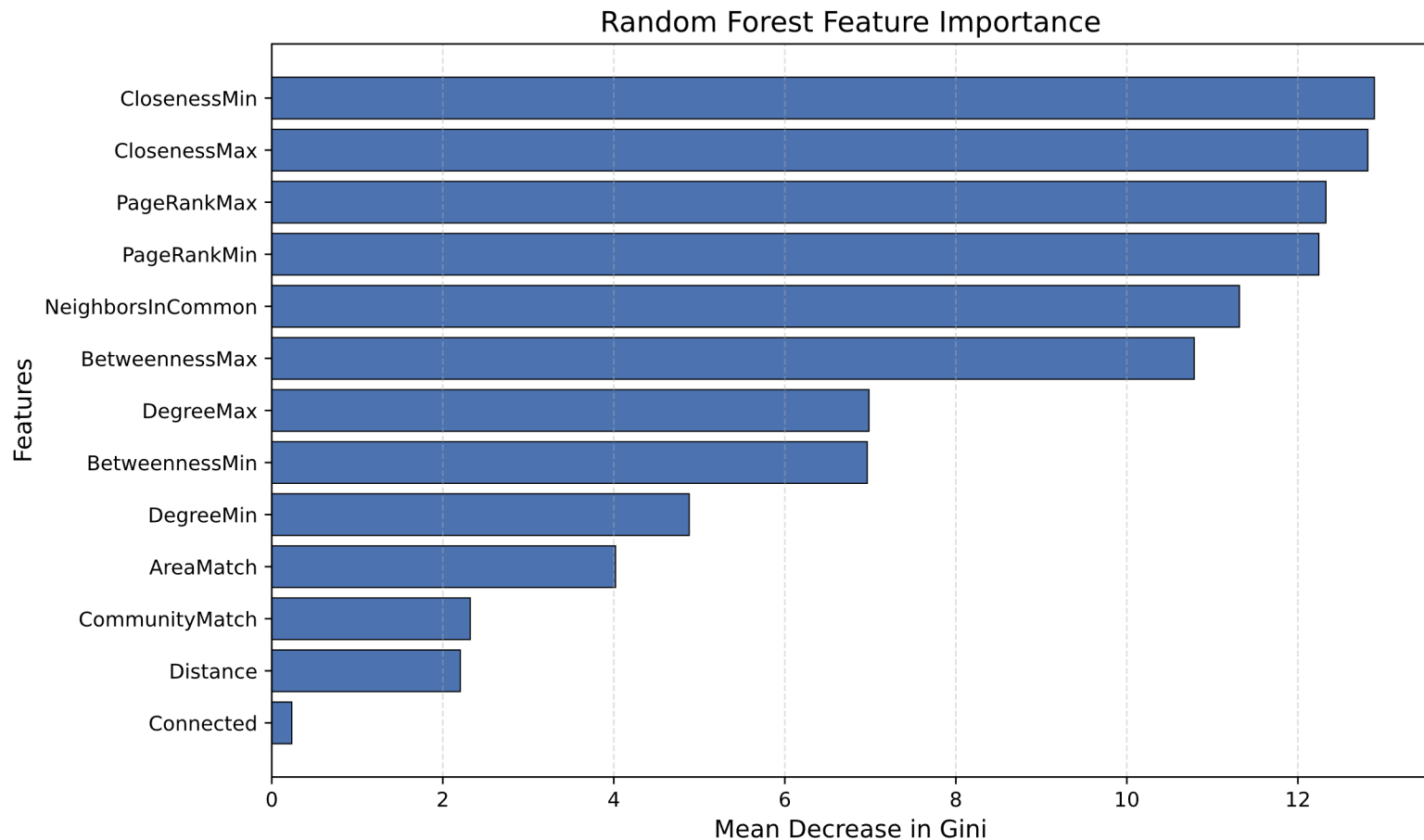
Model Family: Binomial
Link Function: Logit
Method: Maximum Likelihood Estimation (MLE)
No. Observations: 6484
Df Model: 13
Df Residuals: 6470

Coefficients:

	Coef	Std Err	z	P> z	[0.025	0.975]	Signif
Intercept	-4.4146	0.4334	-10.1858	0.0000	-5.2641	-3.5651	***
Connected	6.0301	1.3304	4.5326	0.0000	3.4226	8.6377	***
Distance	-1.2948	0.1963	-6.5950	0.0000	-1.6795	-0.9100	***
PageRankMin	8.4963	33.6003	0.2529	0.8004	-57.3592	74.3517	
PageRankMax	5.0461	13.5111	0.3735	0.7088	-21.4352	31.5274	
BetweennessMin	0.0027	0.0012	2.1860	0.0288	0.0003	0.0051	*
BetweennessMax	-0.0004	0.0006	-0.6337	0.5263	-0.0015	0.0008	
DegreeMin	0.0659	0.0605	1.0891	0.2761	-0.0527	0.1844	
DegreeMax	0.0101	0.0385	0.2616	0.7936	-0.0655	0.0856	
ClosenessMin	-9.4347	3.7488	-2.5167	0.0118	-16.7822	-2.0872	*
ClosenessMax	0.4107	1.6111	0.2549	0.7988	-2.7470	3.5685	
NeighborsInCommon	2.4145	0.8097	2.9821	0.0029	0.8276	4.0014	**
CommunityMatch	0.2796	0.2329	1.2007	0.2299	-0.1768	0.7360	
AreaMatch	1.0653	0.1669	6.3830	0.0000	0.7382	1.3923	***

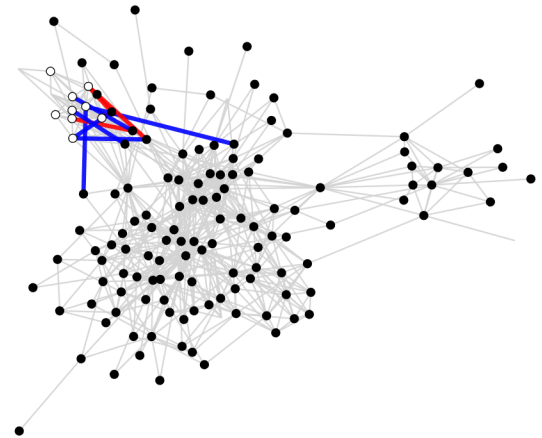
Significance codes: '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1

Random Forest Variable Importance



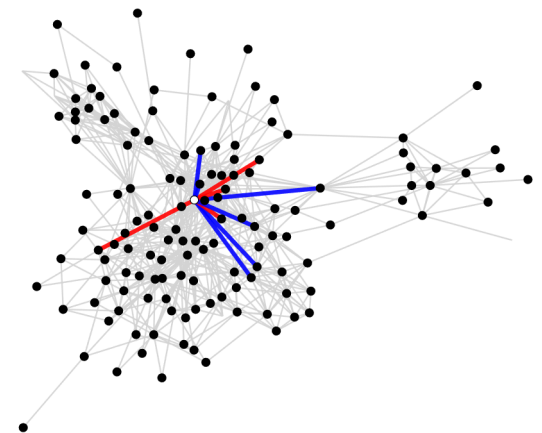
Connecting Isolated Communities

- How can we connect the pipeline team to the rest of the network?
- Flag the 10 edges with the highest predicted link probability in 2001
 - Predicted probabilities: 11–25%
 - 4/10 links actually appeared in 2001
 - Other edges might have appeared with internal social networking



Friend Recommendations

- How can we better connect John Lavorato (CEO) to his employees?
- Flag the 10 individuals with highest predicted link probability in 2001
 - Predicted probabilities: 10-51%
 - 4/10 links actually appeared in 2001
 - Other edges might have appeared with appropriate interventions

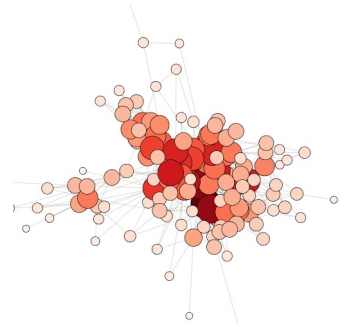


Conclusion

Social Networks: The Analytics Edge

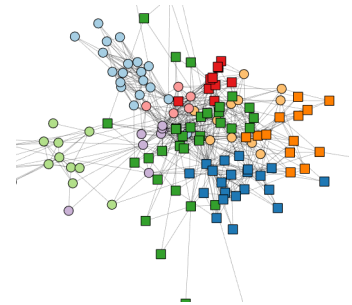
Centrality

- Quantifies the importance / prominence of nodes in a network
 - [Several similar, yet different, metrics]



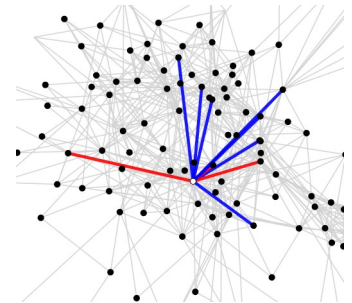
Community detection

- Identifies communities in a network based on existing interactions
 - [One among many techniques]



Link prediction

- Identifies most “promising” new links to strengthen the network
 - Predictions on *dynamic* networks



Use of Social Network Analytics

- Formal organization structure is not necessarily optimized to foster communication/collaboration between business units
- Community detection identifies who is currently collaborating
 - Community detection is a power descriptive tool on network dynamics but does not give us the “right” interventions
- Organizations may benefit from catalyzing communications between relevant business units
 - **Internal social network analysis:** important focus of HR consulting
- Link prediction allows us to identify how “natural” it might be for certain connections to be formed between units/persons

About me

Professor Peng Sun

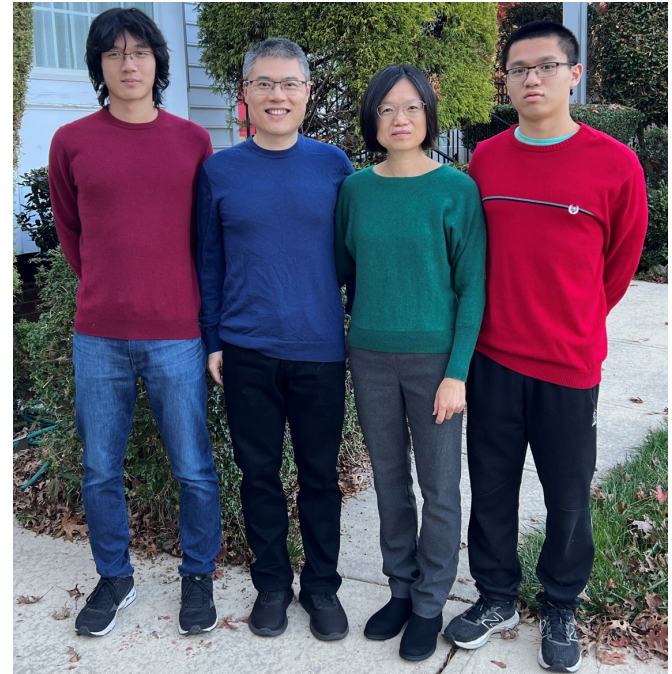
JB Fuqua Professor

Courses Taught

- Decision Models
- Strategic Modeling and Business Dynamics
- Dynamic Programming and Control (PhD course)

Interests

- Decision making under uncertainty
- Optimization models for incentive management
- Healthcare analytics
- Sustainability
- Running
- Movies
- Documentaries on nature, science and history



Coordinates

- psun@duke.edu
- W325

End Slide



I'm looking forward to engaging with
you this term.

See you Monday!